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Domains of life affected by stigma towards leprosy-affected people in Indonesia, the drivers, manifestations, and consequences: A qualitative systematic review

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TITLE PAGE

Domains of life affected by stigma towards leprosy-affected people in Indonesia, the drivers, manifestations, and consequences: A qualitative systematic review

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Abstract

Objective: To provide knowledge about the domains of life affected by stigma on leprosy affected persons in Indonesia, the manifestations of stigma, the driving factors and consequences.
Design: Qualitative systematic review.
Study selection: PubMed, CINAHL, ProQuest, Taylor and Francis, and Google Scholar were used to systematically search original studies that used a qualitative design in the last 20 years. English and Indonesian language were applied to the search. Quality assessment and thematic synthesis were applied.
Data extraction: Information relating to the study characteristics, stigma domains and types following Weiss Extended Scambler's Hidden Stress Model, stigma consequences, and drivers were extracted.
Results: Of the 3184 studies, 37 manuscripts were reviewed. Seven themes were identified. Three themes showed community, domestic, and intimate relationship as the impacted private domains. Four public domains were health, economic, education, and public entitlements. Enacted stigma was mainly discussed in studies, compared to the anticipated and internalized stigma. Ten stigma driving factors were found, ranging from the negative and positive concepts linked with the condition up to aspects not relatable to the disease process. Five areas of consequences were shown. Impacts on public rights, such as education were still very minimally explored, although school absence was very often mentioned. Stigma manifestation, drivers, and consequences in most of the public domains were still least explored.
Conclusion: Leprosy-affected persons in Indonesia experienced and felt stigma from private to public domains of life. Broad aspects, including what is specifically attached to the culture and history of a particular region, are linked with stigma manifestations. The approach in one domain seems to affect another domain. More exploratory studies are needed in the endemic areas outside Java, especially considering the number of studies that are still lacking there and the unique culture in each region of Indonesia.

Keywords
leprosy, stigma, Indonesia, qualitative, review

Strength and limitations of this study

- This review is a response to the basic concept of stigma, which appreciates the importance of local perspectives to generate appropriate stigma knowledge.
- This study complements the language limitation of previous reviews.
- Aspects linked and not linked with the disease process are influencing stigma manifestation.
- Stigma felt and experienced in each life domain appear to be interrelated.
- Current knowledge about the manifestations, determinants, and consequences of stigma towards leprosy-affected people in education and public facilities domains, and non-Java provinces is scarce.

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Introduction

Even though the treatment has been found decades ago and the possibility of being cured is wide open[1], to live as a leprosy-affected person (LAP) has never been easy.[2] Stigma and leprosy are strongly attached. Studies have shown that the life burden of the affected persons is greatly influenced by the stigma rather than the disease itself. Stigma would not only suffer the people infected, but also their affiliation.[3] This is due to leprosy for long has been perceived as a contagious disease, a form of violation of moral or divine instruction[4] and a debilitating disease marked by paralysation and devaluation of one's social role and social identity.[5] These concepts still exist nowadays, therefore, experiences of being rejected, subtly or harshly excluded and inability to disclose can still be found in the recent literature.[2,4,6–9]

In Indonesia, leprosy is known as a disease with a long history and still a burden up until now. Since hundreds of years ago, leprosy has affected all five big islands in Indonesia.[10] Even though a significant improvement in leprosy treatment in Indonesia has made the leprosy cases decline gradually, in the last 10 years, the decline in the number of LAPs is slow.[11] According to the latest report, Indonesia has the third-highest number of leprosy cases globally, after India and Brazil[12]. There are 10 out of 32 provinces and 142 districts/cities that have not yet reached elimination in Indonesia.[11,13] One of the reasons for this is stigma and discrimination.[13]

The knowledge stigma towards LAP in the specific context of Indonesia is needed. Jones *et al* and Yang *et al* emphasized that stigma would be understood through a look at the local world of what is disliked and unwanted in the social setting.[14,15] Knowing that Indonesia is rich with cultures, reflected in the 300 ethnic groups spread across 17,504 islands and 32 provinces[16] and the fact that the spread of leprosy is not only centralised in a specific part of the country, there must be various concepts pertaining on the stigma related to leprosy in the country. Reviews on this were hard to find, even more in the international literature.[8] In addition, the recent reviews in the area of stigma and leprosy also did not include the Indonesian language in the search and seemed to generalize findings from different contexts, mainly from English written literature.[3,17] This review is aimed at providing this knowledge. This study identified, synthesized, and discussed qualitative literature in the field of leprosy in Indonesia. Domains of LAP's life penetrated by stigma in Indonesia would be explored, as well as the stigma manifestations, drivers, and consequences in each of the domains.

Method

Literature search

Eligibility criteria

The inclusion criteria were studies conducted in Indonesia and published in Indonesian and English languages. Qualitative studies and mixed methods studies with a qualitative component were included. Only studies that used primary data and of which the full-text copy online was available, were incorporated. Publication types included were scholarly article journals, dissertations or theses, and manuscripts that resided in the university's repositories in Indonesia, published in 20 years (January 2000 to December 2020).

Search strategy

This review complies with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[18] Studies were searched in multiple electronic databases: PubMed, CINAHL, Taylor and Francis, ProQuest Databases, and Google Scholar (see supplement material 1). Search terms used stigma, leprosy, driver, manifestations, and consequences and their synonyms as keywords.

Study selection

Firstly, all titles and abstracts of the retrieved literature were screened to remove duplicates (YM). Furthermore, studies that were not in line with the eligibility criteria were removed. In the next step, the quality of studies was appraised using the Critical Appraisal Skills Programme (CASP) tool for qualitative research.[19] Studies that did not address more than four criteria were not included in the synthesis. Discrepancies between researchers(YM, ER, AG) were resolved through discussion.

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Data extraction and analysis

Extraction and synthesis of the data were conducted. The following characteristics were extracted: type of study design and method, study perspective, number of participants, age range, and study location. Extraction also included variables reflecting the review domain: the type of stigma (internalized, anticipated, or enacted) following Weiss (2008) Extended Scambler's Hidden Stress Model, the setting of stigma, the consequences of stigma, and the drivers of stigma.[20] Thematic synthesis was used to analyse the data[21]. Each step was done manually by YM and checked by ER and AG. Final themes derived from the review were then discussed and agreed upon between reviewers (YM, ER).

Patient and Public Involvement

Not applicable as it was not appropriate or possible to involve patients or the public in the design, conduct, reporting, or dissemination plans for a review study.

Ethical statement

Not applicable. This study is a review of articles.

RESULTS

Search results

The literature finding (Figure 1) resulted in 3184 studies, of which 2375 were left after removal of duplication. From the titles and abstract screening, 2088 studies were excluded. A full-text assessment for eligibility was conducted for 287 articles. The remaining articles for CASP appraisal in total were 47 studies, including 4 other eligible articles from serendipitous findings. After the CASP appraisal(see supplement material 2), 37 studies were left for synthesis.

Description of the included studies

Among the 37 studies (Table 1), the majority were published in the past five years from 2015 to 2020(n=28). Most of them were qualitative(n= 34) and three were mixed methods. Only a few(n=17) stated the design used clearly. Phenomenology was used the most(n=9). Others were case studies(n=6), comparative ethnographic(n=1) and life history(n=1) respectively. The majority of studies took place in Java Island(n=29). Only a limited number of studies were conducted from the provinces outside Java i.e., South Sulawesi (n=5) and one study each from North Maluku, North Sumatera, and West Papua provinces. Some studies did not provide detailed information about the frequency of data collection, yet, at least 560 interviews and 27 times focus groups were identified from the reviewed literature. 21 studies were published in Indonesian language and 16 studies in English.

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Table 1 Description of studies included in the review

Research (first author, year)	Province(s)	City/district(s)	Study design, type	Data collection method(s)	Study perspective(s)	Age criteria of LAP included
Al Bana, 2014[22]	East Java	Mojokerto	Life history, qualitative	Interviews	LAP	Adult
Arisal, 2020[23]	South Sulawesi	Makassar	Design not stated, qualitative	Interviews, observation and documentation	LAP, family, and community	Adolescence
Armajin, 2019[24]	North Maluku	Ternate	Design not stated, qualitative	Interviews	LAP	Adult
Aulia, 2019[25]	East Java	Mojokerto	Design not stated, qualitative	Interviews and observation	LAP and health worker	Age not clearly addressed, possibly adult
Dadun, 2016[26]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family, community, religious leader, health worker, social and business worker	Age not clearly addressed, possibly adult
Dary, 2017[27]	Central Java	Jepara	Phenomenology, qualitative	Interviews	LAP, health worker	Adult, adolescence
Fajar, 2002[28]	East Java	Gresik	Design not stated, mixed-methods	Interviews	LAP	Adult
Hidayat, 2020[29]	East Java	Sumenep	Phenomenology, qualitative	Interviews	Community	Adult
Huda, 2011[30]	East Java	Sampang	Design not stated, qualitative	Interviews and observation	LAP, community, health worker	Adult
Indow, 2019[31]	West Papua	Manokwari	Case study, qualitative	Interviews	LAP	Adult
Jatimi, 2020[32]	East Java	Sumenep	Phenomenology, qualitative	Interviews	LAP	Adult
Jufriyanto, 2020[33]	East Java	Pamekasan	Phenomenology, qualitative	Interviews	LAP	Adult
Kusharnanto, 2013[34]	Central Java	Blora	Design not stated, qualitative	Interviews, observation and documentation	LAP	Adult
Lusli, 2015[35]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus groups	LAP	Age not clearly addressed, possibly adult
Lusli, 2016[36]	West Java	Cirebon	Design not stated, mixed-methods	Interviews at baseline and post	LAP	Age not clearly addressed, possibly adult
Lusli, 2017[37]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family (caregiver), counsellors	Age not clearly addressed, possibly adult
Marbaits, 2012[38]	East Java	Gresik	Case study, qualitative	Interviews and observation	LAP and family	Adult
Nainggolan, 2017[39]	Banten	Tangerang	Case study, qualitative	Interviews and documentation	LAP, community, health worker, and social worker	Adult
Najmuddin, 2013[40]	South Sulawesi	Makassar	Phenomenology, qualitative	Interviews	LAP	Age not clearly addressed, possibly adult
Peters, 2013[8]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family, community, religious leader, health worker	Adult, children
Peters, 2014[7]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adult
Peters, 2016[41]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adult
Pribadi, 2018[42]	East Java	Pasuruan	Case study, qualitative	Interviews, observation and documentation	LAP, family, and community	Adult
Rai , 2020[43]	West Java	Cirebon	Design not stated, mixed-methods	Interviews	LAP	Adolescence, adult
Rai , 2020[44]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adolescence, adult
Romadhon, 2020[45]	East Java	Sampang and Mojokerto	Comparative ethnographic, qualitative	Interviews and observation	LAP, family, community, health worker	Age not clearly addressed, possibly adult
Saing, 2016[46]	South Sulawesi	Palopo	Design not stated, qualitative	Interviews and observation	LAP, community, health worker, government worker	Age not clearly addressed, possibly adult
Sandi, 2018[47]	East Java	Ngawi	Phenomenology, qualitative	Interviews	LAP	Adult
Schuller, 2010[48]	South Sulawesi	Gowa and Maros	Design not stated, qualitative	Interviews and focus group	LAP, community, and religious leader	Adult
Siregar, 2018[49]	West Java	Depok	Phenomenology, qualitative	Interviews and observation	Family	Adult
Sodik, 2016[50]	East Java	Trenggalek	Design not stated, qualitative	Interviews and observation	LAP, family, community, organization (co-worker)	Adult
Susanto, 2017[51]	East Java	Jember	Phenomenology, qualitative	Interviews	LAP	Adult
Tarigan, 2019[52]	North Sumatera	Karo	Design not stated, qualitative	Interviews, observation and documentation	LAP	Adult
Verayanti, 2016[53]	South Sulawesi	Makassar	Case study, qualitative	Interviews, observation and documentation	LAP, family, community, health worker	Adolescence, adult
Wahyuni, 2017[54]	East Java	Sumberglagah	Case study, qualitative	Interviews	LAP	Adult
Widodo, 2018[55]	East Java	Nganjuk	Design not stated, qualitative	Interviews, observation and documentation	LAP, community, health worker	Adult
Yudanagara, 2020[56]	Central Java	Jepara	Phenomenology, qualitative	Interviews	LAP	Adult

Table 2 Domains, manifestations, consequences, and drivers of stigma towards LAPs in Indonesia

Research (first author, year)	Domain(s) where stigma impacts*	Stigma manifestation(s)	Stigma consequence(s) on LAPs	Stigma driver(s)
Al Bana, 2014[22]	C, D, IR, H, PE	Enacted, Anticipated, Internalized	Social behaviour, Psychological	Fear of contagion, Physical factors
Arisal, 2020[23]	C, D, IR, EC	Enacted, Internalized	Social behaviour, Psychological, Occupational/Financial	Social background, Incapability performing gender role
Armajin, 2019[24]	C, D, IR, EC, ED, PE	Enacted, Anticipated, Internalized	Social behavioural, Psychological, Health	Physical factors, Fear of contagion
Aulia, 2019[25]	H	Enacted, Internalized	Psychological, Health	Fear of contagion, Social and health assistance factors, Physical factors, Religious and/or mystical
Dadun, 2016[26]	EC, IR, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial	Social background, Fear of contagion, Physical factors, Religious and/or mystical reason
Dary, 2017[27]	C, D, IR	Enacted, Anticipated	Social behaviour, Psychological	Religious and/or mystical reason, Fear of contagion, Physical factors
Fajar, 2002[28]	D, IR, C, ED, PE	Enacted, Anticipated	Health	Local connotation for LAP, Religious and/or mystical reason, Physical factors, Social and health assistance factors
Hidayat, 2020[29]	C	Enacted	No consequence mentioned	Fear of contagion, Physical factors, Hereditary factor, Religious and/or mystical reason
Huda, 2011[30]	C, IR	Enacted, Internalized	No consequence mentioned	Hereditary factor, Beliefs around prognosis, Fear of contagion, Religious and/or mystical reason
Indow, 2019[31]	D, C	Enacted, Anticipated, Internalized	Psychological	Physical factors, Fear of contagion
Jatimi, 2020[32]	D, C	Internalized	Social behaviour, Psychological	Inability to perform social role, Physical factors
Jufriyanto, 2020[33]	C, D	Enacted	No consequence mentioned	Fear of contagion, Physical factors
Kusharnanto, 2013[34]	C, EC ED, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Education	Physical factors, Fear of contagion
Lusli, 2015[35]	C, D EC, H	Enacted, Anticipated, Internalized	Social behaviour, Health, Psychological, Occupational/Financial	Physical factors, Fear of contagion
Lusli, 2016[36]	C, D, IR, EC, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health, Occupational/Financial	Physical factors, Incapability to perform social role, Social and health assistance factors
Lusli, 2017[37]	D, IR, EC, ED	Enacted, Internalized	Social behaviour, Psychological Health, Occupational/Financial, Education	Hereditary factor, Religious and/or mystical reason, Beliefs around prognosis, Incapability to perform social role, Fear of contagion, Social background
Marbaits, 2012[38]	C, H	Anticipated, Internalized	Social behaviour, Psychological, Health	Social and health assistance factors, Hereditary factor, Beliefs around prognosis
Nainggolan, 2017[39]	C, D, IR, EC, H	Enacted, Internalized	Social behaviour, Occupational/Financial, Psychological	Religious and/or mystical reason, Physical factors, Social and health assistance factors
Najmuddin, 2013[40]	C, H, EC	Anticipated, Internalized	Psychological, Health, Occupational/Financial	Physical factors, Incapability to perform social role, Local connotation for LAP, Hereditary factor, Fear of contagion, Religious and/or mystical reason, Beliefs around prognosis
Peters, 2013[8]	D, C, EC, ED, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological Occupational/Financial, Health, Education	Physical factors, Social and health assistance factors, Fear of contagion
Peters, 2014[7]	D, C, EC	Anticipated, Internalized	Social behaviour, Psychological, Health	Physical factors, Fear of contagion
Peters, 2016[41]	C, D, EC	Enacted, Anticipated	Social behaviour, Psychological, Occupational/Financial	Physical factors
Pribadi, 2018[42]	C, EC, PE	Enacted, Internalized	Social behaviour, Psychological, Health	Physical factors
Rai , 2020[43]	C, H	Enacted, Internalized	Psychological	Religious and/or mystical reason, Physical factors, Social background
Rai , 2020[44]	C, D, H	Enacted, Anticipated, Internalized	Social behaviour, Health	Religious and/or mystical reason, Fear of contagion, Physical factors
Romadhon, 2020[45]	C, EC, H	Enacted, Anticipated, Internalized	Social behaviour, Health	Residential area
Saing, 2016[46]	C	Enacted	No consequence mentioned	Physical factors, Fear of contagion, Beliefs around prognosis
Sandi, 2018[47]	D, C, IR, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health	Physical factors, Fear of contagion
Schuller, 2010[48]	C, D, IR	Enacted, Anticipated, Internalized	Social behaviour, Psychological	Incapability to perform social role, Physical factors, Religious and/or mystical reason, Hereditary factor, Local connotation for LAP
Siregar, 2018[49]	D, C, EC, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Health	Fear of contagion
Sodik, 2016[50]	C	Enacted	Social behaviour, Psychological	Religious and/or mystical reason, Fear of contagion
Susanto, 2017[51]	C	Enacted	Social behaviour	No drivers mentioned
Tarigan, 2019[52]	C, D, EC, H, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Health, Education	Physical factors, Incapability to perform social role
Verayanti, 2016[53]	D, C, EC	Enacted, Internalized	Social behaviour, Psychological, Occupational/Financial	Local connation for LAP, Physical factors
Wahyuni, 2017[54]	C, D, IR, EC, H, PE, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health, Education	Physical factors, Fear of contagion, Residential area
Widodo, 2018[55]	C, EC	Enacted	Social behaviour, Occupational/Financial	Fear of contagion, Beliefs around prognosis
Yudanagara, 2020[56]	C, D, EC, PE	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial	Religious and/or mystical reason, Physical factors, Fear of contagion

**Community (C):* relationship with neighbours, friends, community members; *Domestic(D):* impacts in household affairs, family roles, relations with family member and relatives; *Intimate relationship (IR):* issues around spouse or romantic relationship; *Health (H)*: healthcare worker approach, ambience in health facility, effect of disfigurement for healthcare; *Economic(EC):* income-related aspects, co-worker relationship; *Education (ED):* responses from school society; *Public entitlements (PE)*: access to and acceptance in public transport, market, kiosk, and worship place.

Synthesis

The domains of life penetrated by stigma, stigma manifestations, drivers, and consequences towards LAP in Indonesia were synthesized (Table 2). Seven domains impacted by stigma were 'community', 'domestic', and 'intimate relationship', 'health', 'economic', 'education', and 'public entitlements' domains. Seven themes were developed to explain stigma manifestations, drivers and consequences in each domain. The description of stigma manifestations (see supplement material 3), ten categories of stigma drivers on LAP in Indonesia (see supplement material 4) and five areas of stigma consequences were also elaborated (see supplement material 5).

Stigma in private domains

Community: difficult relationship and alterations in social standing

LAP in Indonesia faced many difficulties to have a satisfactory relationship in the community, indicating an alteration in their social standing within the society. Issues were experienced with neighbours[8,27,33,56] or others in the hometown,[27] friends,[49,53] and not only limited to adult members.[47,54] Enacted,[2,8,33,34,39,40,43–45,47–49,22,50–52,54–56,23,26–31] anticipated,[2,7,44,47,49,52,54,56,8,22,24,27,28,34,38,40] and internalized stigma,[2,7,51–53,56,8,23,30,36,42,47–49] were linked to this.

Forms of enacted stigma were avoidance to meet,[22,28,35,37,40,41,54] to visit or to receive LAP as a house guest,[8,30,50,56] to greet,[47] to talk with,[24,29,40,55] to be physically near,[22,29,36,44,49] or to have contact with,[8,31,50,55] and even to look at them.[56] Rejections and hesitations towards LAP were experienced in an un-replied smile and hand offer,[2,8,47] an insincere handshake gesture,[56] a recitation of prayer to the food made by LAP,[30] the disapproval to live in the same neighbourhood,[27,30,39,45] to return to hometown,[27] and to be in the same body of water e.g. river.[54] Exclusions were mentioned.[23,28,36,40,47,50,53,54] These include the restriction to participate in the community,[35] and the absence of invitations to the events or gatherings.[48,54,55] LAP also experienced expulsion,[29] the suggestion to leave the community,[27] gossips,[26,37,54] the verbal mocks,[26,28,34,44,47,51–54] and non-verbal insults such as people covered nose and spat when they passed the house of LAP.[56]

The drivers of this experienced stigma were associated with physical factors,[2,22,54,29,31,33,43,44,46,52,53] fear of contagion,[30,31,33,47,55] religious or mystical reasons,[30,44,50,56] local connotation on 'leprosy' word that labels the disease as fatal, ugly, insulting, or should be ignored,[28,48] living in the residential area known for leprosy,[45] health worker visits to the home,[8,36] and an inadequate social rehabilitation after released from treatment.[39] Conditions unrelated to leprosy, such as a poorer financial condition,[43] being a woman or a part of the minority in ethnic or religion in the social setting were also linked to unfair treatments.[23]

The anticipated stigma among LAP were the fear of avoidance,[2,7,8,24,38,44] rejection,[24,37,54] exclusion,[24,44] insult,[2,7,8] rumours,[7] and irritating others if initiating the conversation[56]. Limited studies mentioned the reasons for this anticipated stigma, i.e. changes in physical feature,[36,54] hereditary aspect,[40] beliefs around leprosy such as being contagious, a curse, a spell, a trial, or a sin,[40] and the negative past experiences in the community.[56]

The internalized stigma include self-labelling, self-deprecation, and ashamed of social life. LAP perceived themselves as disgusting,[54] dirty,[36] dangerous,[36] contagious,[54] only an object of charity,[48] incurable,[8] different,[32] inferior,[23] would receive no friendly manner,[8] and an enemy of the community.[30] LAP in this stage could feel people afraid of them,[56] awkward gathering with others,[52] and self-retraction from the community is a reasonable move.[7] The most repeated reasons for the internalized stigma were the physical appearance.[36,42,53] Others were belief in disease transmission[7] and witnessed avoidance by the healthcare worker(HCW).[8]

Stigma in the community domain had impacted mental health, social behaviour, physical, and the financial status of LAP. Psychologically, LAP experienced sadness,[35,36,47,53,56] stress,[38,56] hurt,[34,53,56] blaming self,[36] loneliness,[27,36] easy to be irritated,[42] feeling unsafe in the community,[23] inability to see the future,[36] loss of passion to live,[26] having suicidal thought[7,36,38,56] and act[38]. These effects could linger long in LAP.[8] At the social behavioural level, LAP faced difficulty to make friends[2,56] and to practice meaningful local greetings

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gestures,[2,55] isolated themselves,[2,8,54,27,36,38,42,44,47,51,53] reluctant to share emotional burdens with others[2] or to be close with new people,[56] lacked motivation for social activities,[8] passive at home,[2] reduced activities outside[8,43,47,52] or if they did it otherwise, concealed it from the family.[41] If LAP went for activities outside, they would hide the deformities with hands hidden under the armpits, wear long pants, long sleeves, hijabs, or hats.[24,54] Due to stigma, LAP's social standing and reputation declined in the community.[44] LAP might choose or be suggested by the family to leave the neighbourhood and move to another community that would accept them.[34,39,53] Stigma in the community domain also risked physical health. LAP concealed their disease status,[2,7,28,40,44,49] even from their family members,[41] denied the signs,[28,38] lied about the disease,[24,49] did self-mutilation,[7] rejected HCW's visits to the home,[36] hesitated or secretly sought treatment,[24,28] and fear to the health centre.[28]

Domestic: devaluation of status, roles, and control over own body

The diminished social status and roles of LAP occurred in the family. This was depicted in enacted,[8,27,53,54,56,28,33,37,39,44,47–49] anticipated,[7,8,22,24,31,36,48,49] as well internalized stigma.[2,24,32,37,48,52,54] Studies noted the feeling of embarrassment to have LAP as a family member and tried to conceal their diagnoses.[28,44,53] LAP also received unpleasant treatment from their children.[28,33,54] Children hesitated to perform the *'Salim'* tradition (a form of respectful greeting to parents or older people in some regions of Indonesia) to affected parents,[54] requested parents to hide whenever peers or friends were visiting the house,[54] and stop attending school because of shame to have an affected-parent.[28] In the house, LAP experienced negative comments and rejection to take part in household roles[2,36,48] and was not allowed to touch other family members.[7,24,36,37] A not welcoming gesture[28] even though LAP has been declared cured,[39,56] the feeling of fear from family members to talk with LAP,[49] avoidance from members of extended family,[27,49] siblings,[22,47] and in-law[27] also happened. A case of where affected elderly felt no family members present was written.[8] Other manifestations were the exclusion from family events,[48] expulsion from home,[28] and forms of separation that portrayed the decline of autonomy over own body, such as separating cutlery, bed, clothes[28,48] and house,[28,32] reinforcing LAP to clean their belongings,[28,48] and restricting LAP to take a bath.[47] Enacted stigma in the domestic domain were driven by the LAP's physical appearance,[27,54] local long-held connotation about LAP,[28] the concern of blood transmission,[49] ineffective post-treatment rehabilitation for the LAP and their family,[28,39] and community beliefs that leprosy is a curse or sin[27] which then induced the family's fear of bad image due to the presence of LAP.[28]

Anticipated stigma was manifested in the fear being avoided,[24,49] excluded,[24] receiving negative comments,[2,31] and other reactions from the family about the disease.[31] Concern over inability to perform gender roles in the family such as supporting the children,[36] restrictions to touch the child,[7] and the humiliation the family members would receive were mentioned.[48] Almost no studies notified the driving factors for the anticipated stigma in the domestic domain. One reason was surrounding beliefs that leprosy is a curse and cannot be cured[48].

Devaluation of roles and status was also reflected in the forms of internalized stigma. LAP expressed a shameful feeling to be in the family members,[54] self-doubts to carry out the responsibility to the family[37] for example to breastfeed, to touch the child, to cook, or to fulfil a financial need in the family.[7] Blaming self as the family burden or a trouble-maker,[2,32,38,48] not responsible as breadwinner,[52] a source of transmission to the family,[7,24,57] were expressed in the literature. Self-stigma at the domestic level was also shown in the decision to withdraw from other family members and separate eating utensils and clothes to wash.[24] Fear of contagiousness,[24] inability to do gender roles due to decreased performance,[32,48,54] and physical appearance[54] were some reasons found promoting self-stigma in LAP at the domestic level.

Stigma in the domestic domain had mental, social, and health impacts. Mentally, LAP experienced sadness,[8,31,37,39,52,56] frustration,[39,56] disappointment,[52,56] anger,[39] despair[37], loneliness,[27,37] feeling hopeless about life,[39] guilt,[32,48] continuous feelings of shame and could not fully accept oneself.[52] In the social-behaviour level, a presence of 'distance' with other family members was felt.[37] LAP isolated his or her self from a family member,[39] chose not to communicate to the family,[2,49] and moved out to stay elsewhere.[39,53,56] Stigma at the domestic level also brought health risks to the LAP. LAP concealed and lied about the disease

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to the family,[24,31,36] not routinely taking the medicine due to lack of family support,[28] and experienced a decline in personal hygiene.[47]

Intimate relationship: trouble in having a life partner

Enacted,[22–24,27,28,30,37,39,48,54] anticipated,[26,36] and self-stigma[48,54] on LAP affected the chance to have a romantic partner, to marry, and to preserve marriage. Not only for the LAP, but the influence of stigma also could impact the difficulty of LAP's family member to have a life partner.[48]

In the enacted form, studies noted family rejections on LAP's romantic relationship. Disapproval to marry a person who was affected,[39] who was not affected,[22] or who has or had a history of leprosy in the family line[28,30,48] were mentioned. A marriage plan cancellation was found[30,39] even after "cured" had been declared officially.[39] Romantic relation was rejected by previous partners[39] and by youngsters outside the village[54]. Amongst LAP who were married, avoidance from a spouse,[27,37] a suggestion to divorce from the family members including in-laws and spouses after diagnoses,[28,36] and an expulsion of child-in-law affected with the disease occurred.[28] Reasons were beliefs that leprosy is inherited,[29,30,40,48] a curse,[27] local connotations that pictured leprosy as ugly, terrible, or an insulting disease,[28] the fear that the family lineage will be affected,[28,30] the effort to protect the family good name,[30] and concern of contagiousness from the infected spouse.[22,27] A study noted that the problem to finding a life partner might be bigger for a woman than a man, as a result of a local stereotype that an affected woman could not serve their husband and the household affairs optimally.[23]

Fearing a romantic partner will leave and avoid the LAP[26,36] were the anticipated stigma the intimate relationship level. Manifestations of internalized stigma were feeling of shame to build a relationship[48] and no one wants to marry.[54] These feelings were influenced by physical appearance.[48,54]

Issues in intimate relationships resulted in health risks, mental, and social behaviour of the LAP. Studies mentioned several negative emotions such as sadness,[37] feeling stress,[39] despair,[47] being stuck and unfulfilled in life,[39] and suicidal thinking.[24] In social behavioural level, changes in the quality of relationship amongst family members and decided to leave home.[39] Related to the health risk, LAP chose to lie about the disease from their partner.[36]

Stigma in public domain

Health: challenges in healthcare-seeking and disfigurement effect

Stigma also occurred in the health domain, demonstrated by the behaviour of people at the healthcare facilities and LAP's self-response to the disease development. Although not as many as other private domains, some enacted,[8,25,26,44,47,49,54] anticipated,[45] and internalized stigma,[25,34,38,52] were found.

The enacted type of stigma was demonstrated by the healthcare worker (HCW) who refrained from handshaking or other physical contact with the LAP,[8,26,44,54] delayed health service,[26,47] hesitated[25] or rejected to perform the treatment,[25,47] delivered hurtful words[25] and gave an awkward gesture.[25] Not only from HCW, but LAP also experienced being stared at by the visitors at the health facility as well.[49] Prioritizing men to go to the health services than women occurred.[23] Physical appearance,[54] the concern of transmissions,[25,26,49] a lack of sensitiveness in the general health care centre,[25] and a belief that man is capable in taking care of things in daily life than woman affected[23] were written as the driving factors for this experienced stigma. Worrying about the exclusion from the family members could also be a reason why HCW performed a different approach felt unpleasant by the LAP.[26]

An anticipated form of stigma in the health domain was the fear of being associated with the widely stigmatized leprosy disease if visiting a free health event in the local area.[45] There was no clear reason for this fear. However, it was likely to be facilitated by the community collective desire to reject the label that is put to their territory (i.e. Mandangin island) where the free health clinic was held.[45] Through an old story, for generations the area was stigmatized as the 'island for leprosy-affected by many and also the neighbourhood communities.[45]

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In the health domain, physical appearances related to the disease progress and the treatment effects developed internalized stigma amongst the LAP. Disfigurement effects produced a shameful feeling to face HCW or to visit the healthcare facility,[25,34,52] a belief that leprosy cannot be healed,[8,38] and the feeling of 'I am still a patient' after completing the treatment.[40] The problem of appearances also led to stigma occurrences in all other domains.

Stigma in the health domain brought health and psychological implications. The unpleasant act from HCW promoted the feeling that no one would be friendly anymore to the LAP,[8] trauma,[25] developed further negative feelings towards HCW,[25,47] and an absence of confidence in meeting HCW which then made the disease worsened.[25] The feeling of being an 'eternal patient' also could leave LAP with frustration, life pessimism, not seeking treatment,[38] and abandoning personal health.[40]

Economic: a lack of chance and desire to be productive

In the economic domain, enacted,[2,8,52,53,55,23,24,26,36,37,41,42,49] anticipated,[7] and internalized stigma,[2,8,34,39,40,53,56] were related to the difficulty to access job and a declining desire to be productive.

Enacted stigma in the economic domain manifested in the following phenomenon: lack of job opportunities,[23,24,26,47,52] rejection when applying for a job,[2] suggestion to resign from the family members,[8] being fired from a job,[8,47,53] temporary suspension until fully recovered,[42] exclusion and avoidance from co-workers,[37] an awkward expression or an absence of customers at the LAP's kiosk,[8,41,52] and a lower chance of financial support to run a business.[26] A different physical appearance,[2,52] contagiousness concern,[8] aspects related to the organizational policy to get a bank loan,[26] and being a woman[23] were reasons that promoted these experiences. Women affected would have a double burden due to their role in domestic affairs.[23]

In the economic domain, only one anticipated stigma was found which is the fear of being avoided by customers.[7] The internalized stigma was reflected in a declining desire to be productive. LAP sensed they would not be able to find or do any job anymore,[34,56] would not be fitted for a proper job,[39,53] stopped working,[8] and labelling their self fit to be a beggar[40] or an object of charity.[2] Feeling of shame to work at daylight[56] also depicted internalized stigma. These self-stigma forms were promoted by the physical appearance,[2,56] having a disability,[53] decreased physical performance,[8,34] presence of stigma in the community[53] including a long-lived local meaning that associates the LAP with chronic disease, full of the wound, begging and living in poverty.[40]

Stigma had impacted the occupational and financial, social, mental, and health status of LAP. LAP experienced a loss of customers,[8,52] bankruptcy,[37] discontinuity between their current work and their potentials,[53] and a limitation to work (e.g. working only at night) to avoid people noticing them.[56] Unemployment could lead LAP to poverty and criminal behaviour,[26] dependence on external aid,[36,53] including becoming a beggar,[23,53] not prioritizing treatment,[54] worsening health,[42] labelling self as a sick person forever,[2] and returning to the place where they were a patient.[56] Psychologically, LAP felt discouragement,[37,54] self-doubt,[34,39] lack of passion to live,[26] frustration,[36] suicidal thinking and attempt.[24,37] Stigma in the economic domain also made LAP conceal the disease due to fear of losing customers.[7]

Education: constraints in accessing education and developing self

Indonesian children who were infected with leprosy or whose family members had leprosy were reported to stop attending school temporarily or permanently.[8,22,26,28,36–38,40,54] Other children continued their education at school but admitted that they had to endure shame,[53] hid their conditions from school authorities,[54] or hesitated to develop themselves.[23]

The enacted stigma originated from school society were insults,[8,36,47,54] people who took a distance,[8,37] gossip about the student's parents that contracted with leprosy,[24] or a school bus that stopped far from the residential area where many LAPs lived.[54] The knowledge about the drivers of these acts were lacking in the literature. Some reasons mentioned were the fear of contagion and the presence of darkening skin.[54]

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The anticipated stigma included fear of being mocked or avoided by others[8] or LAP parents who feared that their children will be shunned if schoolmates notice their condition.[54] The internalized stigma manifested in the feeling of shame to be present at school.[36,54] No studies mentioned the anticipated and internalized stigma drivers in the education domain.

The intensity of stigma in the education domain could be influenced by factors not directly related to the disease such as limited school options nearby the residential area, which made an infected child could not change school and should bear stigma during the study[54]. The importance of having education in Indonesian society may also influence the force of stigma related to education.[8]

Besides disturbances in school participation, no studies explored more the impacts of stigma towards affected people in the education domain.

Public entitlements: 'distance' to market, public transportation, and worship places

Stigma towards LAP had widened the gap to access public facilities that supports life. LAP experienced unfair treatment at the kiosk, market, public transportation, and worship places. At the kiosk and market, some sellers refused to sell their goods to the LAP[22,54] and LAP experienced being starred by the surrounding people.[54] LAP also felt expulsion from public transportations,[22,28] difficulty to access public transportation that passes their residential area.[54] In the worship places, LAP also felt discrimination[24,28] such as no one wanted to be near.[28] Reasons for these forms of enacted stigma notified by the literature were the physical look,[54] fear of transmission,[54] and the long-lived local connotation attached with the LAP[28].

Amongst LAP, the feeling of shame to be present in public places could occur and reflect the internalization of stigma. For example, studies mentioned embarrassment to show at the prayer places.[42,54] Reasons for this were the physical appearances such as darkened skin[42] and deformity.[54] The footwear that should be removed before praying also facilitated this hesitation, because of the greater possibility, that a deformity in the foot may be seen.[54]

Thinking about ending life was found as a psychological impact of discrimination in public entitlements such as accessing worship places and public transportation.[24] Other than this, consequences of stigma in this domain could not be found.

Coverage of the domains affected by stigma

Supplement 6 shows the coverage of the domains in the reviewed studies. This table reveals how extensive and consistent research had discussed the stigma phenomenon in each domain of life of LAP in Indonesia.

Discussion

This review was aimed to systematically retrieve the qualitative evidences on leprosy-related stigma in Indonesia. Departed from literature, the synthesis gave seven themes on the domains of life penetrated by stigma in Indonesia context, from the community, domestic, intimate relationship, health, economic, education, and public entitlements. These themes represented how far stigma could impact LAP life in Indonesia in the socioecological levels.

The 'community' domain was known as the most salient domain impacted by stigma or the most saturated domain that has been studied qualitatively in Indonesia across settings, although stigma behaviour itself can be dynamic considering the possibility of sociocultural changes. Stigma phenomenon in the 'domestic', 'intimate relationship', 'health', and 'economic' domains had received quite attentions as well. Stigma forms, drivers, and consequences were still least explored under 'education' and 'public facilities' domains, indicating lack of concern in related knowledge and research. Of all type of stigma, internalized and perceived stigma were not as easy to be identified in the literature as the experienced stigma.

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The stigma drivers were not always related to the negative concepts associated with leprosy conditions. Positive behaviours linked with the conditions such as home visits by HCW and other factors which were not directly associated with leprosy for example economic background were also found to induce stigma. These results imply that the causes of stigma can be very broad, including what seems to be considered positive and not relatable to the disease process, thereof all aspects need be taken into account when developing anti-stigma interventions for LAP. In terms of research, this would complement a previous global approach review.[58]

This study recommends that the efforts to reduce stigma related with leprosy in Indonesia, need to touch all of the seven domains. Not only because of the consequences of stigma in each domain were found to be heart-breaking and violating human rights, but also the above results showed that what happened in one domain seemed affecting other domains of life. As an example, the avoidance from HCW (health domain), had promoted an instant feeling of not worthy of a social relationship in LAP(community domain)[8] and the attitude of avoiding health facilities which eventually become a risk factors to stigma occurrence in community interaction.[54]

The above could also implied to an added knowledge into the recent Health and Stigma Discrimination Framework by Stangl *et al* (2019).[59] It can be suggested that the stigmatization process could be both fuel and reinforced not only by stigma manifestations and drivers, but also by stigma consequences or outcomes. Moreover, this would mean that stigma across domains can be interrelated, thus suggesting the anti-stigma programs need to be done simultaneously in all impacted domains to prevent stigma long-list occurrences of stigma.

From the findings, suggestions regarding the focus of anti-stigma program in each domain could be deliberated. In the community life, attention is needed to an open environment for LAP to participate in the local practices meaning for the people, including daily practices that is easy to be unnoticed but valuable such as replying greetings. Involving cultural leaders to restructure the long-held local labels for LAP is needed. In the domestic domain, concerns should be given to improve marital prospect, quality of marital life, role practices in the family, and family cohesiveness. In the public domains, particularly in the area of healthcare services, is how to be best preventing physical disability and improving manners of HCWs in leprosy and general care. HCWs need to know how to communicate bad news and give therapeutic communication to the LAP and their families, throughout the disease course and especially at the point of diagnosis. HCWs should also give supports to the family members as caregivers or key decision maker, of how to take care of and communicate to LAP during and after the treatment phase. Finally, HCWs should make sure the rehabilitation after the completion of therapy is adequate to the needs of LAP. In economic domain, focus should be given to providing job and preventing job loss, including building awareness amongst the employers. Despite the small number of literature found, in education domain, focus is to create a friendly school environment for the LAP and student whose family member is affected. Regarding public entitlements, the provision of access to, acceptance and equal treatment in the market, transportation, and religious places are required.

Qualitative studies in nine out of ten leprosy endemic provinces (list of provinces uses the 2017 data)[13] i.e. North Sulawesi, Southeast Sulawesi, West Sulawesi, Gorontalo, Maluku, North Maluku, West Papua, and Papua are scarce. This suggests the need to increase explorative studies on the living experiences of LAP in population most at risks. This information would be needed to increase a more contextual awareness for the larger public, practitioners, and other researchers. Stigma knowledge in the domains of education and public entitlements in Indonesia are needed. Views and experiences from children and key persons in the structural field both private and government-owned are also essential. In addition, the quality of qualitative literature published in the Indonesian journals also needs to be improved.

Conclusion

LAP in Indonesia have experienced stigma in various life contexts. Stigma against LAP was carried out not only by the wider community but also by their loved ones such as their spouses or other family members.

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37 studies included have described the types, consequences, and factors that encourage stigma in Indonesia. Further research is still needed to capture the stigma felt and experienced by LAP outside Java provinces.

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Figure Legend:

Figure 1 - PRISMA flow diagram depicting the study selection process

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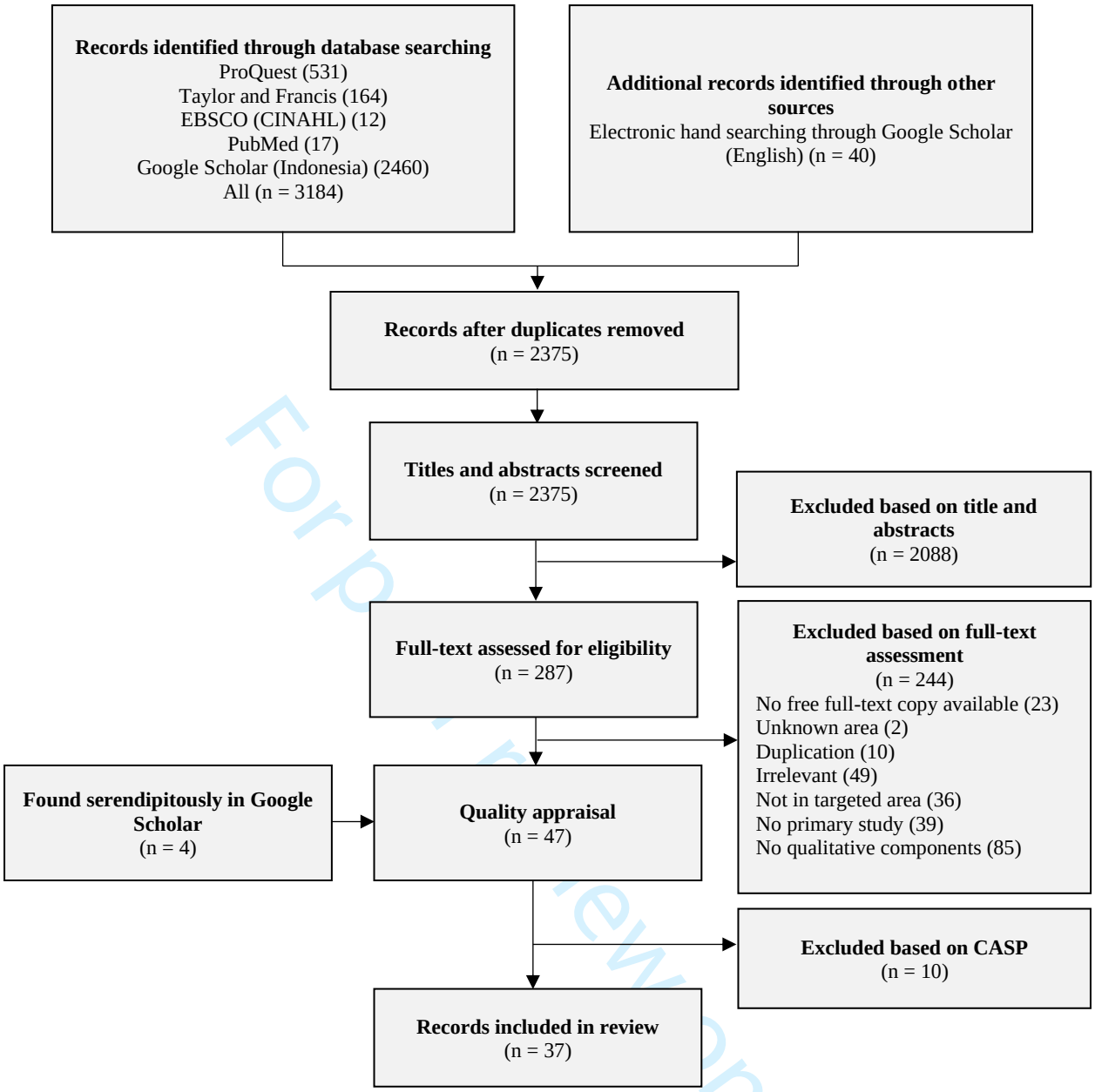


Figure 1. PRISMA flow diagram depicting the study selection process

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Supplement Material 1

Syntaxes for search

Databases	Syntaxes	Results
ProQuest Database	(("risk factor" OR "risk factors") OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR "quality of life" OR manifestation* OR outcome* OR satisfaction) AND (leprosy OR "Hansen's Disease" OR "Hansen Disease" OR leprae) AND (qualitative OR anthropolog* OR cultur* OR "focus group" OR "focus-group" OR ethnography OR "grounded theory" OR interview) AND Indonesia)	531 (Scholarly Journal Article) + 18 (Dissertation /Theses)
Taylor and Francis	(“risk factor*” OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR “quality of life” OR manifestation* OR outcome*) AND (leprosy OR “Hansen's Disease” OR “Hansen Disease” OR leprae) AND Indonesia AND (qualitative OR anthropolog* OR cultur* OR “focus group” OR “focus-group” OR ethnography OR “grounded theory” OR interview)	164 (Scholarly Journal Article)
EBSCO (CINAHL)	(“risk factor*” OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR “quality of life” OR manifestation* OR outcome*) AND (leprosy OR “Hansen's Disease” OR “Hansen Disease” OR leprae) AND Indonesia AND (qualitative OR anthropolog* OR cultur* OR “focus group” OR “focus-group” OR ethnography OR “grounded theory” OR interview)	12 (Scholarly Journal Article)
PubMed	#1“risk factor*”[tw] OR cause*[tw] OR determinant*[tw] OR stressor*[tw] OR belief*[tw] OR construct*[tw] OR practice*[tw] OR driv*[tw] OR facilitat*[tw] #2 impact*[tw] OR effect*[tw] OR influence*[tw] OR result*[tw] OR stigma[tw] OR “quality of life”[tw] OR manifestation*[tw] OR outcome*[tw] OR satisfaction[tw] #4 leprosy[tw] OR “Hansen's Disease”[tw] OR “Hansen Disease”[tw] OR leprae[tw] #5 Indonesia[tw] #6 qualitative[tw] OR anthropolog*[tw] OR cultur*[tw] OR “focus group”[tw] OR “focus-group”[tw] OR ethnography[tw] OR “grounded theory”[tw] OR interview[tw]	17 (Scholarly Journal Article)
Google Scholar (English)	leprosy AND qualitative AND (stigma OR "quality of life") AND ("east java" OR "west java" OR gorontalo OR sulawesi OR papua OR maluku)	40 (Electronic Hand Searching)
Google Scholar (Indonesia)	*kusta AND kualitatif ("faktor risiko" OR penyebab OR determinan OR "sumber daya" OR dampak OR pengaruh OR stigma OR "kualitas hidup" OR sulawesi OR gorontalo OR maluku OR papua OR jawa OR indonesia)*	2460 (All type of publication)

Supplement Material 2

Critical Appraisal Skills Programme (CASP) score for the reviewed studies

No	Study (first author, year)	CASP										
		Was there a clear statement of the aims of the research?	Is qualitative methodology appropriate?	Was the research design appropriate to address the aims of the research?	Was the recruitment strategy appropriate to the aims of the research?	Was the data collected in a way that addressed the research issue?	Has the relationship between researcher and participants been adequately considered?	Have ethical issues been taken into consideration?	Was the data analysis sufficiently rigorous?	Is there a clear statement of findings?	How valuable is the research?	CASP Score
1	Al Bana, 2014[22]	1	1	1	1	1	0	1	1	1	1	9
2	Arisal, 2020[23]	1	1	1	1	Can't tell	1	0	1	1	1	8
3	Armajin, 2019[24]	1	1	1	1	Can't tell	0	0	Can't tell	1	1	6
4	Aulia, 2019[25]	1	1	1	1	Can't tell	1	0	0	1	1	7
5	Dadun, 2016[26]	1	1	1	1	1	0	1	1	1	1	9
6	Dary, 2017[27]	1	1	1	1	Can't tell	0	1	1	1	1	8
7	Fajar, 2002[28]	1	1	Can't tell	1	1	0	0	Can't tell	1	1	6
8	Hidayat, 2020[29]	1	1	1	1	1	0	1	1	1	Can't tell	8
9	Huda, 2011[30]	1	1	1	1	1	0	0	1	1	1	8
10	Indow, 2019[31]	1	1	1	1	1	0	1	1	1	1	9
11	Jatimi, 2020[32]	1	1	1	1	1	0	1	1	Can't tell	1	8
12	Jufriyanto, 2020[33]	1	1	1	1	1	0	1	1	1	1	9
13	Kusharnanto, 2013[34]	1	1	1	1	1	Can't tell	1	1	1	1	9
14	Lusli, 2015[35]	1	1	1	1	1	Can't tell	1	1	1	1	9
15	Lusli, 2016[36]	1	1	1	1	1	1	1	1	1	1	10
16	Lusli, 2017[37]	1	1	1	1	1	1	Can't tell	1	1	1	9
17	Marbaits, 2012[38]	1	1	1	1	1	0	0	1	1	1	8
18	Nainggolan, 2017[39]	1	1	1	1	1	0	0	Can't tell	1	1	7
19	Najmuddin, 2013[40]	1	1	1	1	1	0	1	1	1	1	9
20	Peters, 2013[8]	1	1	1	1	1	Can't tell	1	1	1	1	9
21	Peters, 2014[7]	1	1	1	1	1	1	1	1	1	1	10
22	Peters, 2016[41]	1	1	1	0	1	0	0	Can't tell	1	1	6
23	Pribadi, 2018[42]	1	1	1	1	Can't tell	0	Can't tell	1	1	Can't tell	6
24	Rai , 2020[43]	1	1	1	1	1	1	1	1	1	1	10
25	Rai , 2020[44]	1	1	1	1	1	1	1	1	1	1	10
26	Romadhon, 2020[45]	1	1	1	1	1	1	1	1	1	1	10
27	Saing, 2016[46]	1	1	1	Can't tell	Can't tell	0	0	1	1	1	6
28	Sandi, 2018[47]	1	1	1	1	1	Can't tell	1	1	1	1	9
29	Schuller, 2010[48]	1	1	1	1	1	Can't tell	Can't tell	1	1	1	8
30	Siregar, 2018[49]	1	1	1	1	Can't tell	0	1	1	1	1	8
31	Sodik, 2016[50]	1	1	1	1	0	0	0	0	1	1	6
32	Susanto, 2017[51]	1	1	1	1	1	0	1	1	1	1	9
33	Tarigan, 2019[52]	1	1	1	1	1	0	Cant tell	Cant tell	1	1	7
34	Verayanti, 2016[53]	1	1	1	1	1	0	0	1	1	1	8
35	Wahyuni, 2017[54]	1	1	1	Can't tell	1	0	0	1	1	1	7
36	Widodo, 2018[55]	1	1	1	1	1	0	0	0	1	1	7
37	Yudanagara, 2020[56]	1	1	1	1	1	0	Can't tell	1	1	1	8

Supplement Material 3

Description of stigma manifestations towards LAPs in Indonesia

Enacted
Avoidance[2,8,41,44,47,49–51,53,56,22,26–28,30,33,36,37]
Rejection in greetings, handshake or physical touch[8,26,31,44,47,50,54,56]
Reluctance to talk[29,49,55]
Absences of invitations to gatherings[48,50,54,55]
Restriction to participate[2]
Negative comment, gossip, or mock[24,25,54,26,34,36,43,49,51–53]
Spitting on property and nose was covered[56]
Covering nose[56]
Starring[49]
Rejection to live in the same neighbourhood[45]
Expulsion[28,29]
Unfair treatment due to social background[23,43]
Recitation of a specific prayer to reject transmission[30]
Rejection to marry or to build romantic partner[22,24,30,39,47,48,54]
Acceptance and giving suggestion to divorce[36,48]
Prohibition of touching family member[37]
Concealment of diagnosis by family members[28,44]
The view as a family disgrace[53]
Family rejection even though has been cured[39]
Command not to take a bath[47]
Utensils, clothes, bed separation[28]
Suggestion to leave the house[27]
Shortage of job[24,26,52,55]
Rejection from job[2]
Alienation at work[36,49]
Suspension to work[42]
Dismissal from work[8,53]
Refusal to be in the same body of water (e.g. river, water at rice field) [8,54]
Rejection or hesitation to buy items sold by LAP[8,37,41,52,54]
Hesitation, rejection to help, or delayed treatment from healthcare provider[25,26,47]
Hurtful comments at school[8,34,37,52]
Physical distance[24,28,29,36,44]
Absence of transportation at the residential area where LAP lived[54]
Rejection at the market[22]
Rejection from public transportation driver and passengers[24,28,56]
Anticipated
Fear of avoidance[7,8,24,36,38,41,43,49,52]
Fear of insults, humiliation, or negative comments[2,7,8,38,48,49]
Fear people would reject to meet[7,36,54]
Fear of infecting others[26,27]
Worry people will notice or suspect something wrong[7,24,28,36]
Fear if interactions will make people will feel offended[56]
Worry the deceased body of LAP would not be buried at the public cemetery[22]
Fear family members will afraid of the disease[31]
Fear to be excluded by the family and neighbours[24,31,44]
Worry of being rejected to take care of family[7,36]
Fear romantic partner will leave[26]
Fear difficulty to have a romantic partner[26]
Internalized
Feeling ashamed of self [8,25,54,32,34,36,44,47,49,51,53]
Seeing self as an abnormal or different[2,32,48,54]
Seeing self as dirty, ugly, disgusting, unclean[22,26,36,54,56]
Seeing self as an eternal-patient, will not be healed or normal[2,8,22,37,38,40,43,56]
Seeing self as a danger for other[36]
Feeling no one want to marry[54]
Seeing self a poor person and fit to be a beggar[40]
Seeing self as a family troublemaker or burden[32,37,48]
Seeing self as an object of charity[2,48,54]
Seeing self as having a dangerous disease[40]
Seeing a stamp of 'leprosy' at self[56]
Feeling inferior[8,23]
Feeling people will not be friendly anymore[8]
Feeling as a cursed person even after cured[39]
Feeling unworthy[39]
Feeling will not be able to get a job or work again[34,56]
Feeling people are afraid of self[56]

Supplement Material 4

Description stigma drivers towards LAPs in Indonesia

Drivers	Description	
Physical factors[2,8,32–34,36,39–44,22,46–48,52–54,56,24–29,31]	Physical appearance due to deformity or a darkening skin, unpleasant physical odour, decreased physical function, decreased performance due to physical weakening, continuous pain after cured	Related with the disease progress
Fear of contagion[2,7,37,40,44,46,47,49,50,54,56,8,24,25,27,28,31,33,34]	Fear of fast transmission from breathing the same air, standing in the similar body of water, shaking hands, eating food prepared by the person affected, carrying the deceased body of a person, using the same personal object such as glasses, towels, clothes	
Hereditary factor[29,30,37,38,40,48]	Believed that leprosy can be inherited in the family line	
Local connotation for LAP[28,40,48,53]	Negative connotation attached to the local word directed to leprosy affected persons (e.g *'Nibamparekangji'* in South Sulawesi that has a meaning 'to ignore or treat them like they do not exist or *'Judam'* in East Java that means 'ugly and hated by people')	
Religious and/or mystical reason[25,27,48,50,56,28–30,37,39,40,43,44]	Believed that leprosy is caused by sin, curse, punishment on bad deeds, evil spirit influence, or spell	
Incapability performing social role[23,32,36,37,40,48,52]	Lack or no opportunity to perform previous routine or responsibility in and for the family, continuous failure in relationship, job, school	
Factors related to social and health assistance[8,25,28,36,38,39]	Home visits from health officer*[can be considered positive approach, yet could drive stigma]*, exhaustion in caring for patients, health worker communication style, or an inadequate social rehabilitation	
Beliefs around prognosis[30,37,38,40,46,55]	Believed leprosy as a long-term disease, a high fatality disease, and an incurable disease	
Residential area[45,54]	Believed that the residential area where persons-affected live is a dangerous zone or by living there you will also get stigmatized	
Social background[23,37,43]	Ethnic, religion, gender, or economic background	Related with the disease progress

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Supplement Material 5

Description of stigma consequences on LAPs in Indonesia

Social behaviour
Self-isolation[7,8,49,51,52,54,24,26,27,35,36,38,42,47]
Avoiding new people, home visits, and social activities[8,24,35,36,47,50,56]
Apathy to others[38]
Hiding activities from family members[41]
Difficulty to marry, build romantic relationship or friendship[22,35,39,48,56]
Estranged or damaged relationship with significant others[8,26,37,39]
Being passive at home[35,36,38,47]
Changing place to live[23,27,34,39,52,53,56]
Hiding face or body parts when going outside [24,34,36,54]
Loss of social image[44]
Limited inter-island movement[45]
Being used for political and economic purpose[23,45]
Psychological
Sadness[8,31,54,56,35–37,39,47,49,52,53]
Frustration[8,31,34,36,38,39,47,52,56]
Despair[8,36,47]
Guilt[32,48]
Felling lonely[27,35–37]
Feeling nsafe[24]
Anger or easy to be irritated[39,42]
Inability to be angry[54]
Pessimism[8,36,38,43,53]
Loss of self-confidence[36,47,50]
Feeling unfulfilled[39]
Feeling useless[54]
Being ashamed of self (for a long time period)[8,23,32,41,52,55,56]
Self-disappointment[36,52,54]
Spiritual distress (e.g. questioning God or stopping routine religious activity)[39,47]
Loss of passion to live[26,38,39,41]
Suicidal thinking[7,8,22,24,27,36,38,54,56]
Suicidal attempt[38]
Occupational/financial
Resigning from work[8,36,49]
Difficulty to find job[26,34,35,55]
Losing customer[8,41,52]
Preference to work at night[56]
Feeling shy to find a proper job[34,53]
Weak financial condition[23,26,37]
Dependence on aid to sustain life (e.g. becoming a beggar or leaning on government's aid) [23,26,40,53]
Health
Concealing, lying or denying the diagnoses[2,7,49,8,24,26,31,36,37,40,44]
Avoiding healthcare facility or provider[8,24,25,28,38,40,44,45,54]
Abandoning the sign of leprosy[28,38]
Seeking treatment secretly[28]
Seeking traditional healer[28,44]
Medication noncompliance[28,44]
Lack of family support in treatment[28]
Absence in health education[40]
Unhygienic practice of self-care[47]
Worsening disease[25,42,44]
Weakened physical health[38,42]
Excessive use of drugs[52]
Neglecting general health[40]
Self-mutilation[7]
Education
Stop attending school[8,26,34,36,37,52,54]
Difficulty to access school bus[54]
Having teachers who refuses to live on the same island with the students[45]

Supplement Material 6

Coverage of domain impacted with stigma in the included literature

Themes	Literature																																					
Affected private domains	**Number of studies**	41	30	48	57	32	22	42	36	43	38	55	23	45	2	49	52	47	40	39	8	7	53	50	37	35	34	54	27	31	26	44	33	29	25	28	46	24
Community	35/37	•	•	•	_	•	•	•	•	•	•	•	•	•	•	•	_	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Domestic	22/37	•	•	•	_	_	•	•	_	_	•	•	•	_	•	•	•	_	•	_	•	•	_	_	_	•	_	_	•	•	•	_	_	•	•	•	_	•
Intimate relationship	13/37	•	•	•	_	•	•	•	_	•	_	_	_	_	_	•	•	_	•	_	_	_	_	_	_	_	_	_	•	•	_	_	_	_	_	•	_	_
Affected public domains																																						
Healthcare	15/37	_	_	_	•	•	_	_	_	_	•	•	_	•	_	_	_	•	•	•	•	_	_	_	_	•	•	_	•	-	•	_	•	•	_	•	_	_
Economic	20/37	_	•	•	_	•	_	_	_	_	_	_	_	•	•	•	•	_	•	•	•	•	•	•	_	_	•	_	•	-	•	_	_	•	•	_	•	•
Education	8/37	•	-	•	_	_	_	_	_	_	_	_	_	•	_	•	•	_	_	_	•	_	_	_	_	_	_	_	_	_	_	_	_	•	_	•	_	_
Public facilities	6/37	•	_	•	_	_	_	•	_	_	_	_	_	_	_	_	_	_	_	_	_	_	_	•	_	_	_	_	_	_	_	_	_	_	_	•	_	•

• : mentioned in the literature
- : not mentioned in the literature

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	p.1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	p.2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	p.3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	p.3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	-
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	p.3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	p.3 & suppl.materials
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	suppl. materials
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	p.4.
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	p.4
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	p.3-4 & suppl. materials
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	p.4 & supp. material
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	-
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	p.4

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	p.4 & supp. material 2 (CASP)
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	-
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	p.4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	p.4-7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	suppl. material 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	p.5-6, supp. material 6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	p.7-10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	supp.material 2 (CASP)
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	-
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	p.10-p.11
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	p.11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	p.11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	p.1

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From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Domains of life affected by stigma towards leprosy affected people in Indonesia, the manifestations, drivers, and consequences: a qualitative systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2022-062372.R1
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Date Submitted by the Author:	02-Sep-2022
Complete List of Authors:	Marpaung, Yosi; Universitas Kristen Krida Wacana, Nursing Ernawati, Ernawati; Universitas Kristen Krida Wacana, Nursing Dwivania, Agatha; Universitas Kristen Krida Wacana, Nursing
Primary Subject Heading:	Public health
Secondary Subject Heading:	Infectious diseases, Mental health, Qualitative research
Keywords:	INFECTIOUS DISEASES, Public health < INFECTIOUS DISEASES, QUALITATIVE RESEARCH, MENTAL HEALTH

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TITLE PAGE

Domains of life affected by stigma towards leprosy affected people in Indonesia, the manifestations, drivers, and consequences: a qualitative systematic review

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Abstract

Objective: To provide knowledge about the domains of life of leprosy affected people impacted with stigma, its manifestations, driving factors and consequences in Indonesia.
Design: Qualitative systematic review.
Study selection: PubMed, CINAHL, ProQuest, Taylor&Francis, and Google Scholar were used to systematically search studies with qualitative component that were conducted in Indonesia and published from January 2000 to December 2020 in English or Indonesian language. The search was started in November 2020 and reran in April 2021. Quality assessment and thematic synthesis were applied.
Data extraction: Of the 3184 studies, 37 manuscripts were reviewed. Information relating to the study characteristics, stigma domains and types following Weiss Extended Scambler's Hidden Stress Model, stigma consequences, and drivers were extracted.
Results: Seven themes were identified. Three themes showed community, domestic, and intimate relationship as the impacted private domains. Four were public domains including health, economics, education, and public entitlements. Enacted stigma was mainly discussed in studies, compared to the anticipated and internalized stigma. Ten stigma driving factors were found, ranging from the negative and positive concepts linked with the condition up to aspects not relatable to the disease process. Five areas of consequences were shown. Impacts on public rights, such as education were still very minimally explored, although school absence was often mentioned. Stigma manifestation, drivers, and consequences in most of the public domains were least explored.
Conclusion: Leprosy-affected persons in Indonesia experienced and felt stigma from private to public domains of life. Broad aspects, including what is specifically attached to the culture and history of a particular region, are linked with stigma manifestations. The approach in one domain seems to affect another domain. More exploratory studies are needed in the endemic areas outside Java, especially considering the number of studies that are still lacking there and the unique culture in each region of Indonesia.

Keywords
domain of life, leprosy, stigma, Indonesia, qualitative, review

Strength and limitations of this study

- We include studies written in English and Indonesian language to provide richer information about the specific context of Indonesia and complement previous reviews.
- Findings were derived from various sources including scholarly article journals and academic thesis.
- In text analysis of qualitative studies, differences in meaning with the original author of the article might happen although this risk was minimized by the researchers who were all Indonesians and a consensus to any discrepancies had been attempted.

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Introduction

Even though the treatment has been found decades ago and the possibility of being cured is wide open[1], to live as a leprosy-affected person (LAP) has never been easy.[2] Stigma and leprosy are strongly attached. Studies have shown that the life burden of the affected persons is greatly influenced by the stigma rather than the disease itself. Stigma would not only suffer the people infected, but also their affiliation.[3] This is due to leprosy for long has been perceived as a contagious disease, a form of violation of moral or divine instruction[4] and a debilitating disease marked by paralysation and devaluation of one's social role and social identity.[5] These concepts still exist nowadays, therefore, experiences of being rejected, subtly or harshly excluded and inability to disclose can still be found in the recent literature,[4,6–10] including in Indonesia.[7,8]

In Indonesia, leprosy is known as a disease with a long history and still a burden up until now. Since hundreds of years ago, leprosy has affected all five big islands in Indonesia.[10] Even though a significant improvement in leprosy treatment in Indonesia has made the leprosy cases decline gradually, in the last 10 years, the decline in the number of LAPs is slow.[11] According to the latest report, Indonesia has the third-highest number of leprosy cases globally, after India and Brazil[12]. There are 10 out of 32 provinces and 142 districts/cities that have not yet reached elimination in Indonesia.[11,13] One of the reasons for this is stigma and discrimination.[13]

The knowledge stigma towards LAP in the specific context of Indonesia is needed. Jones *et al* and Yang *et al* emphasized that stigma would be understood through a look at the local world of what is disliked and unwanted in the social setting.[14,15] Knowing that Indonesia is rich with cultures, reflected in the 300 ethnic groups spread across 17,504 islands and 32 provinces[16] and the fact that the spread of leprosy is not only centralised in a specific part of the country, there must be various concepts pertaining on the stigma related to leprosy in the country. Reviews on this were hard to find, even more in the international literature.[8] In addition, the recent reviews in the area of stigma and leprosy also did not include the Indonesian language in the search and seemed to generalize findings from different contexts, mainly from English written literature.[3,17] This review is aimed at fill this knowledge gap. The study identified, synthesized, and discussed qualitative literature in the field of leprosy in Indonesia in order to provide knowledge about the domains of life affected by stigma towards leprosy affected people in Indonesia, including the stigma manifestations in each domain, its driving factors and consequences.

Method

Literature search

Eligibility criteria

The inclusion criteria were qualitative studies and mixed methods studies with a qualitative component conducted in Indonesia and published in Indonesian and English languages. Only studies that used primary data and of which the full-text copy online was available, were incorporated. Publication types included were peer-reviewed scholarly article both from national and international journals and grey literature (i.e. academic theses or dissertations and manuscripts that resided in the university's repositories in Indonesia), published in 20 years (January 2000 to December 2020).

Search strategy

The search was guided by the research question that was formulated with PICo statement: "What are the leprosy affected persons'(Population) domains of life of affected by stigma, including its stigma manifestations, driving factors and consequences(Interest) in Indonesia(Context)." This review complies with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[18] Studies were searched in multiple electronic databases: PubMed, CINAHL, Taylor and Francis, ProQuest Databases, and Google Scholar (see supplement material 1. Syntaxes for search). The use of Google Scholar was made to complement literature findings, since a large number of local studies, written in the Indonesian language, were extremely lacking in other scientific databases. Google Scholar is also connected with a large amount of grey literature we intended to seek, from local universities repositories and local peer-reviewed journals.

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Study selection

Firstly, all titles and abstracts of the retrieved literature were screened to remove duplicates. Furthermore, studies that were not in line with the eligibility criteria were removed. In the next step, the quality of studies was appraised using the Critical Appraisal Skills Programme (CASP) tool for qualitative research.[19] CASP consists of ten questions which discusses three broad issues in the qualitative study: results' validity, quality, and usefulness. Each question comprises of three types of answers in which researchers used a scoring system 1 for "yes", and 0 for "no" and "can't tell". In the table presentation (see supplement 2. Critical Appraisal Skills Programme (CASP) score for the reviewed studies), "can't tell" is written as is to distinguish it from the answer "no". Researchers decided studies that did not address at least six criteria of CASP to not be included in the synthesis. Discrepancies between researchers were resolved through discussion.

Data extraction and analysis

Extraction and synthesis of the data were conducted. The following characteristics were extracted: type of study design and method, study perspective, number of participants, age range, and study location. Extraction also included variables reflecting the review domains: the type of stigma (internalized, anticipated, or enacted) following Weiss (2008) Extended Scambler's Hidden Stress Model, the setting of stigma, the consequences of stigma, and the drivers of stigma.[20] Thematic synthesis was used to analyse these data[21]. The first step was the free line-by-line coding of the text. Review domains mentioned earlier guided this step. Then, the organisation of codes under each review domains were conducted. The codes were grouped into descriptive themes. Finally, we generated analytical themes to emphasize the experience in each domain of life affected by stigma. This was done by reviewing all of the findings under each setting affected. Literature were read repeatedly to ensure all concepts needed to answer research objective were already mapped. Each step was done manually by YMM, commented and checked by ER and ATD. Paper and pen and a humble software (Ms. Excel) were used in the process. Final descriptive and analytical themes derived from the review were then discussed and agreed upon between reviewers (YMM, ER). In addition, a coverage analysis was conducted to search the gap of knowledge in the affected domains of life.

Patient and Public Involvement

Not applicable as it was not appropriate or possible to involve patients or the public in the design, conduct, reporting, or dissemination plans for a review study.

Ethical statement

Not applicable. This study is a review of articles.

RESULTS

Search results

The literature finding (Figure 1) resulted in 3184 studies, of which 2375 were left after removal of duplication. From the titles and abstract screening, 2088 studies were excluded. A full-text assessment for eligibility was conducted for 287 articles. The remaining articles for CASP appraisal in total were 47 studies, including 4 other eligible articles from serendipitous findings. After the CASP appraisal(see supplement material 2), 37 studies were left for synthesis.

Description of the included studies

Among the 37 studies (Table 1), the majority were published in the past five years from 2015 to 2020(n=28). Most of them were qualitative(n= 34) and three were mixed methods. Only a few(n=17) stated the design used clearly. Phenomenology was used the most(n=9). Others were case studies(n=6), comparative ethnographic(n=1) and life history(n=1) respectively. The majority of studies took place in Java Island(n=29). Only a limited number of studies were conducted from the provinces outside Java i.e., South Sulawesi (n=5) and one study each from North Maluku, North Sumatera, and West Papua provinces. Some studies did not provide detailed information about the frequency of data collection, yet, at least 560 interviews and 27 times focus groups were identified from the reviewed literature. 21 studies were published in Indonesian language and 16 studies in English. 26 were journal articles and 11 studies were academic thesis.

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Synthesis

The domains of life penetrated by stigma, stigma manifestations, drivers, and consequences towards LAP in Indonesia were synthesized (Table 2). Seven domains impacted by stigma were 'community', 'domestic', and 'intimate relationship', 'health', 'economic', 'education', and 'public entitlements' domains. Seven themes were developed to explain stigma manifestations, drivers and consequences in each domain. The description of stigma manifestations (see supplement material 3. Description of stigma manifestations towards LAPs in Indonesia), ten categories of stigma drivers on LAP in Indonesia (see supplement material 4. Description stigma drivers towards LAPs in Indonesia) and five areas of stigma consequences were also elaborated (see supplement material 5. Description of stigma consequences on LAPs in Indonesia). Next, the elaboration of findings would be explained narratively to help an easy grasp on the nature of stigma in each domain of LAP's life affected. In each particular domain, stigma manifestation and driving factors would be addressed first, followed by the consequences.

Stigma in private domains

Community: difficult relationship and alterations in social standing

LAP in Indonesia faced many difficulties to have a satisfactory relationship in the community, indicating an alteration in their social standing within the society. Issues were experienced with neighbours[8,22–24] or others in the hometown,[22] friends,[25,26] and not only limited to adult members.[27,28] Enacted,[2,7,8,22,23,26–46] anticipated,[2,7,8,22,24,27,28,32,36,38,42,47–50] and internalized stigma,[2,7,8,23,30,36,42,47–49,51,52] were linked to this.

Stigma manifestation and its driving factors in the community level

Forms of enacted stigma from the community were avoidance to meet,[2,28,33,42,45,47,53] to visit or to receive LAP as a house guest,[8,24,34,48] to greet,[27] to talk with,[36,39,49,53] to be physically near,[26,31,33,40,49] or to have contact with,[8,34,36,52] and even to look at them.[24] Rejections and hesitations towards LAP were experienced in an un-replied smile and hand offer,[2,8,27] an insincere handshake gesture,[24] a recitation of prayer to the food made by LAP,[48] the disapproval to live in the same neighbourhood,[22,30,32,48] to return to hometown,[22] and to be in the same body of water e.g. river.[28] Exclusions were mentioned.[25,27,28,34,37,40,47,53] These include the restriction to participate in the community,[2] and the absence of invitations to the events or gatherings.[28,36,46] LAP also experienced expulsion,[49] the suggestion to leave the community,[22] gossips,[28,29,45] the verbal mocks,[25,27–29,31,35,47,51,54] and non-verbal insults such as people covered nose and spat when they passed the house of LAP.[24]

The drivers of these enacted stigma were associated with physical factors,[2,23,25,28,31,33,43,49–52] fear of contagion,[23,27,36,48,52] religious or mystical reasons,[24,31,34,48] local connotation on 'leprosy' word that labels the disease as fatal, ugly, insulting, or should be ignored,[46,47] living in the residential area known for leprosy,[30] health worker visits to the home,[8,40] and an inadequate social rehabilitation after released from treatment.[32] Conditions unrelated to leprosy, such as a poorer financial condition,[50] being a woman or a part of the minority in ethnic or religion in the social setting were also linked to unfair treatments.[37]

The anticipated stigma felt among LAP in the community domain were fear of avoidance,[2,7,8,31,38,39] rejection,[28,39,45] exclusion,[31,39] insult,[2,7,8] rumours,[7] and irritating others if initiating the conversation[24]. Limited studies mentioned the reasons for this anticipated stigma, i.e. changes in physical feature,[28,40] hereditary aspect,[53] beliefs around leprosy such as being contagious, a curse, a spell, a trial, or a sin,[53] and the negative past experiences in the community.[24]

As for the internalized stigma, self-labelling, self-deprecation, and ashamed of social life. LAP perceived themselves as disgusting,[28] dirty,[40] dangerous,[40] contagious,[28] only an object of charity,[46] incurable,[8] different,[44] inferior,[37] would receive no friendly manner,[8] and an enemy of the community were mentioned.[48] LAP in this stage could feel people afraid of them,[24] awkward gathering with others,[51] and think a self-retraction from the community as a reasonable move.[7] The most repeated reason for the internalized stigma was the physical appearance.[25,40,41] Others were belief about disease transmission[7] and witnessed a healthcare worker(HCW) avoiding.[8]

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Table 1 Description of studies included in the review

Research (first author, year)	Province(s)	City/district(s)	Study design, type	Data collection method(s)	Study perspective(s)	Age criteria of LAP included in the research
Al Bana, 2014[33]	East Java	Mojokerto	Life history, qualitative	Interviews	LAP	Adult
Arisal, 2020[37]	South Sulawesi	Makassar	Design not stated, qualitative	Interviews, observation and documentation	LAP, family, and community	Adolescence
Armajin, 2019[39]	North Maluku	Ternate	Design not stated, qualitative	Interviews	LAP	Adult
Aulia, 2019[55]	East Java	Mojokerto	Design not stated, qualitative	Interviews and observation	LAP and health worker	Age not clearly addressed, possibly adult
Dadun, 2016[29]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family, community, religious leader, health worker, social and business worker	Age not clearly addressed, possibly adult
Dary, 2017[22]	Central Java	Jepara	Phenomenology, qualitative	Interviews	LAP, health worker	Adult, adolescence
Fajar, 2002[47]	East Java	Gresik	Design not stated, mixed-methods	Interviews	LAP	Adult
Hidayat, 2020[49]	East Java	Sumenep	Phenomenology, qualitative	Interviews	Community	Adult
Huda, 2011[48]	East Java	Sampang	Design not stated, qualitative	Interviews and observation	LAP, community, health worker	Adult
Indow, 2019[52]	West Papua	Manokwari	Case study, qualitative	Interviews	LAP	Adult
Jatimi, 2020[44]	East Java	Sumenep	Phenomenology, qualitative	Interviews	LAP	Adult
Jufriyanto, 2020[23]	East Java	Pamekasan	Phenomenology, qualitative	Interviews	LAP	Adult
Kusharnanto, 2013[35]	Central Java	Blora	Design not stated, qualitative	Interviews, observation and documentation	LAP	Adult
Lusli, 2015[2]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus groups	LAP	Age not clearly addressed, possibly adult
Lusli, 2016[40]	West Java	Cirebon	Design not stated, mixed-methods	Interviews at baseline and post	LAP	Age not clearly addressed, possibly adult
Lusli, 2017[45]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family (caregiver), counsellors	Age not clearly addressed, possibly adult
Marbaits, 2012[38]	East Java	Gresik	Case study, qualitative	Interviews and observation	LAP and family	Adult
Nainggolan, 2017[32]	Banten	Tangerang	Case study, qualitative	Interviews and documentation	LAP, community, health worker, and social worker	Adult
Najmuddin, 2013[53]	South Sulawesi	Makassar	Phenomenology, qualitative	Interviews	LAP	Age not clearly addressed, possibly adult
Peters, 2013[8]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family, community, religious leader, health worker	Adult, children
Peters, 2014[7]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adult
Peters, 2016[42]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adult
Pribadi, 2018[41]	East Java	Pasuruan	Case study, qualitative	Interviews, observation and documentation	LAP, family, and community	Adult
Rai , 2020[50]	West Java	Cirebon	Design not stated, mixed-methods	Interviews	LAP	Adolescence, adult
Rai , 2020[31]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adolescence, adult
Romadhon, 2020[30]	East Java	Sampang and Mojokerto	Comparative ethnographic, qualitative	Interviews and observation	LAP, family, community, health worker	Age not clearly addressed, possibly adult
Saing, 2016[43]	South Sulawesi	Palopo	Design not stated, qualitative	Interviews and observation	LAP, community, health worker, government worker	Age not clearly addressed, possibly adult
Sandi, 2018[27]	East Java	Ngawi	Phenomenology, qualitative	Interviews	LAP	Adult
Schuller, 2010[46]	South Sulawesi	Gowa and Maros	Design not stated, qualitative	Interviews and focus group	LAP, community, and religious leader	Adult
Siregar, 2018[26]	West Java	Depok	Phenomenology, qualitative	Interviews and observation	Family	Adult
Sodik, 2016[34]	East Java	Trenggalek	Design not stated, qualitative	Interviews and observation	LAP, family, community, organization (co-worker)	Adult
Susanto, 2017[54]	East Java	Jember	Phenomenology, qualitative	Interviews	LAP	Adult
Tarigan, 2019[51]	North Sumatera	Karo	Design not stated, qualitative	Interviews, observation and documentation	LAP	Adult
Verayanti, 2016[25]	South Sulawesi	Makassar	Case study, qualitative	Interviews, observation and documentation	LAP, family, community, health worker	Adolescence, adult
Wahyuni, 2017[28]	East Java	Sumberglagah	Case study, qualitative	Interviews	LAP	Adult
Widodo, 2018[36]	East Java	Nganjuk	Design not stated, qualitative	Interviews, observation and documentation	LAP, community, health worker	Adult
Yudanagara, 2020[24]	Central Java	Jepara	Phenomenology, qualitative	Interviews	LAP	Adult

Table 2 Domains, manifestations, consequences, and drivers of stigma towards LAPs in Indonesia

Research (first author, year)	Domain(s) where stigma impacts*	Stigma manifestation(s)	Stigma consequence(s) on LAPs	Stigma driver(s)
Al Bana, 2014[33]	C, D, IR, H, PE	Enacted, Anticipated, Internalized	Social behaviour, Psychological	Fear of contagion, Physical factors
Arisal, 2020[37]	C, D, IR, EC	Enacted, Internalized	Social behaviour, Psychological, Occupational/Financial	Social background, Incapability performing gender role
Armajin, 2019[39]	C, D, IR, EC, ED, PE	Enacted, Anticipated, Internalized	Social behavioural, Psychological, Health	Physical factors, Fear of contagion
Aulia, 2019[55]	H	Enacted, Internalized	Psychological, Health	Fear of contagion, Social and health assistance factors, Physical factors, Religious and/or mystical
Dadun, 2016[29]	EC, IR, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial	Social background, Fear of contagion, Physical factors, Religious and/or mystical reason
Dary, 2017[22]	C, D, IR	Enacted, Anticipated	Social behaviour, Psychological	Religious and/or mystical reason, Fear of contagion, Physical factors
Fajar, 2002[47]	D, IR, C, ED, PE	Enacted, Anticipated	Health	Local connotation for LAP, Religious and/or mystical reason, Physical factors, Social and health assistance factors
Hidayat, 2020[49]	C	Enacted	No consequence mentioned	Fear of contagion, Physical factors, Hereditary factor, Religious and/or mystical reason
Huda, 2011[48]	C, IR	Enacted, Internalized	No consequence mentioned	Hereditary factor, Beliefs around prognosis, Fear of contagion, Religious and/or mystical reason
Indow, 2019[52]	D, C	Enacted, Anticipated, Internalized	Psychological	Physical factors, Fear of contagion
Jatimi, 2020[44]	D, C	Internalized	Social behaviour, Psychological	Inability to perform social role, Physical factors
Jufriyanto, 2020[23]	C, D	Enacted	No consequence mentioned	Fear of contagion, Physical factors
Kusharnanto, 2013[35]	C, EC ED, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Education	Physical factors, Fear of contagion
Lusli, 2015[2]	C, D EC, H	Enacted, Anticipated, Internalized	Social behaviour, Health, Psychological, Occupational/Financial	Physical factors, Fear of contagion
Lusli, 2016[40]	C, D, IR, EC, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health, Occupational/Financial	Physical factors, Incapability to perform social role, Social and health assistance factors
Lusli, 2017[45]	D, IR, EC, ED	Enacted, Internalized	Social behaviour, Psychological Health, Occupational/Financial, Education	Hereditary factor, Religious and/or mystical reason, Beliefs around prognosis, Incapability to perform social role, Fear of contagion, Social background
Marbaits, 2012[38]	C, H	Anticipated, Internalized	Social behaviour, Psychological, Health	Social and health assistance factors, Hereditary factor, Beliefs around prognosis
Nainggolan, 2017[32]	C, D, IR, EC, H	Enacted, Internalized	Social behaviour, Occupational/Financial, Psychological	Religious and/or mystical reason, Physical factors, Social and health assistance factors
Najmuddin, 2013[53]	C, H, EC	Anticipated, Internalized	Psychological, Health, Occupational/Financial	Physical factors, Incapability to perform social role, Local connotation for LAP, Hereditary factor, Fear of contagion, Religious and/or mystical reason, Beliefs around prognosis
Peters, 2013[8]	D, C, EC, ED, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological Occupational/Financial, Health, Education	Physical factors, Social and health assistance factors, Fear of contagion
Peters, 2014[7]	D, C, EC	Anticipated, Internalized	Social behaviour, Psychological, Health	Physical factors, Fear of contagion
Peters, 2016[42]	C, D, EC	Enacted, Anticipated	Social behaviour, Psychological, Occupational/Financial	Physical factors
Pribadi, 2018[41]	C, EC, PE	Enacted, Internalized	Social behaviour, Psychological, Health	Physical factors
Rai , 2020[50]	C, H	Enacted, Internalized	Psychological	Religious and/or mystical reason, Physical factors, Social background
Rai , 2020[31]	C, D, H	Enacted, Anticipated, Internalized	Social behaviour, Health	Religious and/or mystical reason, Fear of contagion, Physical factors
Romadhon, 2020[30]	C, EC, H	Enacted, Anticipated, Internalized	Social behaviour, Health	Residential area
Saing, 2016[43]	C	Enacted	No consequence mentioned	Physical factors, Fear of contagion, Beliefs around prognosis
Sandi, 2018[27]	D, C, IR, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health	Physical factors, Fear of contagion
Schuller, 2010[46]	C, D, IR	Enacted, Anticipated, Internalized	Social behaviour, Psychological	Incapability to perform social role, Physical factors, Religious and/or mystical reason, Hereditary factor, Local connotation for LAP
Siregar, 2018[26]	D, C, EC, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Health	Fear of contagion
Sodik, 2016[34]	C	Enacted	Social behaviour, Psychological	Religious and/or mystical reason, Fear of contagion
Susanto, 2017[54]	C	Enacted	Social behaviour	No drivers mentioned
Tarigan, 2019[51]	C, D, EC, H, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Health, Education	Physical factors, Incapability to perform social role
Verayanti, 2016[25]	D, C, EC	Enacted, Internalized	Social behaviour, Psychological, Occupational/Financial	Local connation for LAP, Physical factors
Wahyuni, 2017[28]	C, D, IR, EC, H, PE, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health, Education	Physical factors, Fear of contagion, Residential area
Widodo, 2018[36]	C, EC	Enacted	Social behaviour, Occupational/Financial	Fear of contagion, Beliefs around prognosis
Yudanagara, 2020[24]	C, D, EC, PE	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial	Religious and/or mystical reason, Physical factors, Fear of contagion

**Community (C):* relationship with neighbours, friends, community members; *Domestic(D):* impacts in household affairs, family roles, relations with family member and relatives; *Intimate relationship (IR):* issues around spouse or romantic relationship; *Health (H)*: healthcare worker approach, ambience in health facility, effect of disfigurement for healthcare; *Economic(EC):* income-related aspects, co-worker relationship; *Education (ED):* responses from school society; *Public entitlements (PE)*: access to and acceptance in public transport, market, kiosk, and worship place.

Stigma consequences in the community level

Stigma in the community domain had impacted mental health, social behaviour, physical, and the financial status of LAP. Psychologically, LAP experienced sadness,[2,24,25,27,40] stress,[24,38] hurt,[24,25,35] blaming self,[40] loneliness,[22,40] easy to be irritated,[41] feeling unsafe in the community,[37] inability to see the future,[40] loss of passion to live,[29] having suicidal thought[7,24,38,40] and act[38]. These effects could linger long in LAP.[8] At the social behavioural level, LAP faced difficulty to make friends[2,24] and to practice meaningful local greetings gestures,[2,36] isolated themselves,[2,8,22,25,27,28,31,38,40,41,54] reluctant to share emotional burdens with others[2] or to be close with new people,[24] lacked motivation for social activities,[8] passive at home, [2] reduced activities outside[8,27,50,51] or if they did it otherwise, concealed it from the family.[42] If LAP went for activities outside, they would hide the deformities with hands hidden under the armpits, wear long pants, long sleeves, hijabs, or hats.[28,39] Due to stigma, LAP's social standing and reputation declined in the community.[31] LAP might choose or be suggested by the family to leave the neighbourhood and move to another community that would accept them.[25,32,35] Stigma in the community domain also risked physical health. LAP concealed their disease status,[2,7,26,31,47,53] even from their family members,[42] denied the signs,[38,47] lied about the disease,[26,39] did self-mutilation,[7] rejected HCW's visits to the home,[40] hesitated or secretly sought treatment,[39,47] and fear to the health centre.[47]

Domestic: devaluation of status, roles, and control over own body

The diminished social status and roles of LAP occurred in the family. This was depicted in enacted,[7,8,27,28,33,37,39,44,47–49,53,54] anticipated,[7,8,26,33,39,40,46,52] as well internalized stigma.[2,28,39,44–46,51]

Stigma manifestation and its driving factors in the domestic level

Studies noted the feeling of embarrassment to have LAP as a family member and tried to conceal their diagnoses.[25,31,47] LAP also received unpleasant treatment from their children.[23,28,47] Children hesitated to perform the *'Salim'* tradition (a form of respectful greeting to parents or older people in some regions of Indonesia) to affected parents,[28] requested parents to hide whenever peers or friends were visiting the house,[28] and stop attending school because of shame to have an affected-parent.[47] In the house, LAP experienced negative comments and rejection to take part in household roles[2,40,46] and was not allowed to touch other family members.[7,39,40,45] A not welcoming gesture[47] even though LAP has been declared cured,[24,32] the feeling of fear from family members to talk with LAP,[26] avoidance from members of extended family,[22,26] siblings,[27,33] and in-law[22] also happened. A case where the affected elderly felt no family members present was written.[8] Other manifestations were the exclusion from family events,[46] expulsion from home,[47] and forms of separation that portrayed the decline of autonomy over own body, such as separating cutlery, bed, clothes[46,47] and house,[44,47] reinforcing LAP to clean their belongings,[46,47] and restricting LAP to take a bath.[27] Enacted stigma in the domestic domain were driven by the LAP's physical appearance,[22,28] local long-held connotation about LAP,[47] the concern of blood transmission,[26] ineffective post-treatment rehabilitation for the LAP and their family,[32,47] and community beliefs that leprosy is a curse or sin[22] which then induced the family's fear of bad image due to the presence of LAP.[47]

Anticipated stigma in the domestic domain manifested in the fear being avoided,[26,39] excluded,[39] receiving negative comments,[2,52] and other reactions from the family about the disease.[52] Concern over the inability to perform gender roles in the family such as supporting the children,[40] restrictions to touch the child,[7] and the humiliation the family members would receive were mentioned.[46] Almost no studies discussed the driving factors for the anticipated stigma in the domestic domain. A reason was found around beliefs that leprosy is a curse and cannot be cured[46].

Devaluation of roles and status in the domestic level was also reflected in the forms of internalized stigma. LAP expressed a shameful feeling to be in the family members,[28] a self-doubt to carry out the responsibility to the family[45] e.g. to breastfeed, to touch the child, to cook, or to fulfil a financial need in the family.[7] Blaming self as the family burden or a trouble-maker,[2,38,44,46] not responsible as breadwinner,[51] a source of transmission to the family,[7,39] were also expressed in the literature. Self-stigma was also shown in the decision to withdraw from other family members and to separate eating utensils and clothes to wash.[39] Fear

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of contagiousness,[39] inability to do gender roles due to decreased performance,[28,44,46] and physical appearance[28] were found promoting the previous behaviours.

Stigma consequences in the domestic level

Stigma in the domestic domain caused mental, social, and health impacts. Mentally, LAP experienced sadness,[8,24,32,45,51,52] frustration,[24,32] disappointment,[24,51] anger,[32] despair[45], loneliness,[22,45] feeling hopeless about life,[32] guilt,[44,46] continuous feelings of shame.[51] In the social-behaviour level, a presence of 'distance' with other family members was felt.[45] LAP isolated his or her self from a family member,[32] chose not to communicate to them,[2,26] or moved elsewhere.[24,25,32] Stigma at this domain also brought health risks. LAP concealed and lied about the disease to the family,[39,40,52] not routinely taking the medicine,[47] and declined in personal hygiene.[27]

Intimate relationship: trouble in having a life partner

Enacted,[22,28,32,33,37,39,45–48] anticipated,[29,40] and internalized stigma[28,46] affected LAP's chances to have a romantic partner, to marry, or to preserve marriage. The influence of stigma also could impact the difficulty of LAP's family members to have a life partner.[46]

Stigma manifestation and its driving factors in the intimate relationship level

In the enacted form, studies noted family rejections on LAP's romantic relationship. Disapproval to marry a person who is affected,[32] who is not affected,[33] or who has or had a history of leprosy in the family line[46–48] were mentioned. A marriage plan cancellation was found[32,48] even after declared 'cured' officially.[32] Romantic relationship was ended by previous partner[32] and rejected by fellows outside the village[28]. Amongst LAP who were married, avoidance from a spouse,[22,45] a suggestion to divorce from the family members including in-laws and spouses after diagnoses,[40,47] and an expulsion of affected child-in-law occurred.[47] Reasons to these behaviour were varied from beliefs that leprosy is inherited,[46,48,49,53] a curse,[22] the local connotations that pictured leprosy as an ugly, terrible, or insulting disease,[47] the fear of bringing bad things to the family lineage,[47,48] the effort to protect the family good name,[48] up to concern over contagiousness from the infected spouse.[22,33] A study noted that the problem to find a life partner might be bigger for a woman than a man, as a result of a local stereotype that an affected woman could not serve their husband and the household affairs optimally.[37]

Lack of studies discussed the anticipated and the internalized stigma within this domain. Fearing the romantic partner would leave or avoid were occurred as the anticipated stigma[29,40]. Internalized stigma include a feeling of shame to build a relationship[46] and that no one wants to marry.[28] This feeling was told to be influenced by the physical appearance.[28,46]

Stigma consequences in the intimate relationship level

Issues in the intimate relationships resulted in health, mental, and social risks. Studies mentioned negative emotions such as sadness,[45] feeling stress,[32] despair,[27] being stuck or unfulfilled in life,[32] and suicidal thinking.[39] In the social life, changes was occurred in the quality of relationship amongst family members and a decision to leave the house.[32] Risks in health might also occur as LAP lied about the disease from their partner.[40]

Stigma in public domain

Health: challenges in healthcare-seeking and disfigurement effect

Stigma also occurred in the health domain, demonstrated by the behaviour of people at the health facilities and LAP's response to the disease development. Some enacted,[8,26–29,31,55] anticipated,[30] and internalized stigma,[35,38,51,55] were found.

Stigma manifestation and its driving factors in the health level

The enacted type of stigma was demonstrated among HCWs who refrained handshaking or other physical contact with LAP,[8,28,29,31] delayed health service,[27,29] hesitated[55] or rejected to perform the treatment,[27,55] delivered hurtful words and gave an awkward gesture.[55] Not only from HCW, but LAP also experienced being stared at by the visitors at the health facility as well.[26] Prioritizing men to go to the health services

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than women also occurred.[37] Physical appearance,[28] the concern of transmissions,[26,29,55] a lack of sensitiveness in the general health care centre,[55] and a belief that man is capable in taking care of things in daily life than woman affected[37] were written as the driving factors for enacted stigma. Worrying the negative reaction from family members could also be a reason why HCW performed a different approach felt unpleasant by the LAP.[29]

An anticipated form of stigma in the health domain was the fear of being associated with the widely stigmatized leprosy disease if visiting a free health event in the local area.[30] There was no clear reason for this fear. However, it was likely to be facilitated by the community collective desire to reject the label that is put to their territory (i.e. Mandangin island) where the free health clinic was held.[30] Through an old story, the area was for generations known by many including the surrounding communities as the 'island for leprosy-affected'.[30]

In the health domain, physical appearances related to the disease progress and the treatment effects developed internalized stigma amongst the LAP. Disfigurement effects produced a shameful feeling to face HCW or to visit the healthcare facility,[35,51,55] a belief that leprosy cannot be healed,[8,38] and a feel of being an 'eternal patient' after completing the treatment.[53] The problem of appearances also led to stigma occurrences in all other domains.

Stigma consequences in the health level

Stigma in the health domain brought health and psychological implications. The unpleasant act from HCW promoted the feeling that no one would be friendly anymore to the LAP,[8] trauma,[55] developed further negative feelings towards HCW,[27,55] and an absence of confidence in meeting HCW which then made the disease worsened.[55] The feeling of being an 'eternal patient' also could leave LAP with frustration, life pessimism, not seeking treatment,[38] and abandoning personal health.[53]

Economic: a lack of chance and desire to be productive

In the economic domain, enacted,[2,8,23,24,26,36,37,41,42,49,52,53,55] anticipated,[7] and internalized stigma,[2,8,24,25,32,35,53] were related to the difficulty to access job and a declining desire to be productive.

Stigma manifestation and its driving factors in the economic level

Enacted stigma in the economic domain manifested in the lack of job opportunities,[27,29,37,39,51] rejection when applying for a job,[2] suggestion to resign from the family members,[8] being fired from a job,[8,25,27] temporary suspension until fully recovered,[41] exclusion and avoidance from co-workers,[45] an awkward expression or an absence of customers at the LAP's kiosk,[8,42,51] and a lower chance of financial support to run a business.[29] Physical appearance,[2,51] contagiousness concern,[8] aspects related to the organizational policy to get a bank loan,[29] and being a woman[37] were reasons to these experiences. Women affected would have a double burden due to their role in domestic affairs.[37]

In the economic domain, only one anticipated stigma was found which is the fear of being avoided by customers.[7] The internalized stigma was reflected in a declining desire to be productive. LAP sensed they would not be able to find or do any job anymore,[24,35] would not be fitted for a proper job,[25,32] stopped working,[8] and labelling their self fit to be a beggar[53] or an object of charity.[2] Feeling of shame to work at daylight[24] also depicted internalized stigma. These self-stigma were promoted by the physical appearance,[2,24] disability,[25] a decreased physical performance,[8,35] presence of stigma in the community[25] including a long-lived local meaning that associates the LAP with chronic disease, full of the wound, begging and living in poverty.[53]

Stigma consequences in the economic level

Stigma had impacted the occupational and financial, social, mental, and health status of LAP. LAP experienced a loss of customers,[8,51] bankruptcy,[45] a mismatch between their current work and their potential,[25] and a limitation to work (e.g. working only at night) in order to avoid people noticing.[24] Unemployment could lead LAP to poverty and criminal behaviour,[29] dependence on external aid,[25,40] begging,[25,37] not prioritizing treatment,[28] worsening health,[41] labelling self as a patient for life,[2] and deciding

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to live in the place where they were once a patient.[24] Psychologically, LAP felt discouragement,[28,45] self-doubt,[32,35] lack of passion to live,[29] frustration,[40] suicidal thinking and attempt.[39,45] Stigma in the economic domain also made LAP conceal the disease due to fear of losing customers.[7]

Education: constraints in accessing education and developing self

Stigma manifestation and its driving factors in the education level

In the education domain, the enacted stigma emerged from the school society. These include insults,[8,27,28,40] people who took a distance,[8,45] gossip about the student's parents that contracted with leprosy,[39] or a school bus that stopped far from the residential area where many LAPs lived.[28] The drivers of these acts were lacking in the literature. Some reasons mentioned were the fear of contagion and the presence of a darkening skin.[28]

LAP felt some anticipated stigma. These were fear of being mocked or avoided by others[8] or feared that their child will be shunned if schoolmate finds out about the illness.[28] The internalized stigma in the education domain manifested in the feeling of shame to attend school.[28,40] No studies clearly mentioned what drives these behaviours in the education domain. However, it could be noted that the intensity of these felt stigma could be potentially influenced by a limited school options near the residential area where LAP lived[28]. This had made an infected child could not change school in the midst of inconvenience and should bear stigma throughout his/her education[28]. The importance of having education for the Indonesian society may also influence the force of stigma related to education.[8]

Stigma consequences in the education level

Indonesian children who were infected with leprosy or whose family members had leprosy were reported to stop attending school temporarily or permanently.[8,28,29,33,38,40,45,47,53] Other children continued their education at school but admitted that they had to endure shame,[25] hid their conditions from school authorities,[28] or hesitated to develop themselves.[37] Besides disturbances in school participation, no studies explored more the impacts of stigma towards affected people in the education domain.

Public entitlements: 'distance' to market, public transportation, and worship places

Stigma manifestation and its driving factors in the public entitlements level

Stigma towards LAP had widened the gap to access public facilities that supports life. LAP experienced unfair treatment at the kiosk, market, public transportation, and worship places. At the kiosk and market, some sellers refused to sell their goods to the LAP[28,33] and LAP experienced being starred by the surrounding people.[28] LAP also felt expulsion from public transportations,[33,47] difficulty to access public transportation that passes their residential area.[28] In the worship places, LAP also felt discrimination[39,47] such as no one wanted to be near.[47] Reasons for these forms of enacted stigma notified by the literature were the physical look,[28] fear of transmission,[28] and the long-lived local connotation attached with the LAP[47].

The feeling of shame to be present in public places could occur and reflect the internalization of stigma. For example, studies mentioned embarrassment to show at the prayer places.[28,41] Reasons for this were the physical appearances such as darkened skin[41] and deformity.[28] The footwear that should be removed before praying also facilitated this hesitation, because of the greater possibility that a deformity in the foot might be seen.[28]

Stigma consequences in the public entitlements level

Thinking about ending life was found as a psychological impact of discrimination in public entitlements such as accessing worship places and public transportation.[39] Other than this, consequences of stigma in this domain could not be found.

Coverage of the domains affected by stigma

Supplement 6 (Coverage of domain impacted with stigma in the included literature) shows the coverage of the domains in the reviewed studies. This table reveals how extensive and consistent research had discussed the stigma phenomenon in each domain of life of LAP in Indonesia.

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Discussion

This review was aimed to systematically retrieve the qualitative evidences on leprosy-related stigma in Indonesia. Departed from literature, the synthesis gave seven themes on the domains of life penetrated by stigma in Indonesia context, from the community, domestic, intimate relationship, health, economic, education, and public entitlements. These themes represented how far stigma could impact LAP life in Indonesia in the socioecological levels.

The 'community' domain was known as the most salient domain impacted by stigma or the most saturated domain that has been studied qualitatively in Indonesia across settings, although stigma behaviour itself can be dynamic considering the possibility of sociocultural changes. Stigma phenomenon in the 'domestic', 'intimate relationship', 'health', and 'economic' domains had received quite attentions as well. Stigma forms, drivers, and consequences were still least explored under 'education' and 'public facilities' domains, indicating lack of concern in related knowledge and research. Of all type of stigma, internalized and perceived stigma were not as easy to be identified in the literature as the experienced stigma.

The stigma drivers were not always related to the negative concepts associated with leprosy conditions. Positive behaviours linked with the conditions such as home visits by HCW and other factors which were not directly associated with leprosy for example economic background were also found to induce stigma. These results imply that the causes of stigma can be very broad, including what seems to be considered positive and not relatable to the disease process, thereof all aspects need be taken into account when developing anti-stigma interventions for LAP. In terms of research, this would complement a previous global approach review.[56]

This study recommends that the efforts to reduce stigma related with leprosy in Indonesia, need to touch all of the seven domains. Not only because of the consequences of stigma in each domain were found to be heart-breaking and violating human rights, but also the above results showed that what happened in one domain seemed affecting other domains of life. As an example, the avoidance from HCW (health domain), had promoted an instant feeling of not worthy of a social relationship in LAP(community domain)[8] and the attitude of avoiding health facilities which eventually become a risk factors to stigma occurrence in community interaction.[28]

The above could also implied to an added knowledge into the recent Health and Stigma Discrimination Framework by Stangl *et al* (2019).[57] It can be suggested that the stigmatization process could be both fuel and reinforced not only by stigma manifestations and drivers, but also by stigma consequences or outcomes. Moreover, this would mean that stigma across domains can be interrelated, thus suggesting the anti-stigma programs need to be done simultaneously in all impacted domains to prevent stigma long-list occurrences of stigma.

From the findings, suggestions regarding the focus of anti-stigma program in each domain could be deliberated. In the community life, attention is needed to an open environment for LAP to participate in the local practices meaning for the people, including daily practices that is easy to be unnoticed but valuable such as replying greetings. Involving cultural leaders to restructure the long-held local labels for LAP is needed. In the domestic domain, concerns should be given to improve marital prospect, quality of marital life, role practices in the family, and family cohesiveness. In the public domains, particularly in the area of healthcare services, is how to be best preventing physical disability and improving manners of HCWs in leprosy and general care. HCWs need to know how to communicate bad news and give therapeutic communication to the LAP and their families, throughout the disease course and especially at the point of diagnosis. HCWs should also give supports to the family members as caregivers or key decision maker, on how to take care of and communicate to LAP during and after the treatment phase. Finally, HCWs should make sure the rehabilitation after the completion of therapy is adequate to the needs of LAP. In economic domain, focus should be given to provide job and prevent job loss, including building awareness amongst the employers. Despite the small number of literature found, in education domain, focus is to create a friendly school environment for the LAP and student whose family member is affected. Regarding public

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entitlements, the provision of access to, acceptance and equal treatment in the market, transportation, and religious places are required.

Qualitative studies in nine out of ten leprosy endemic provinces (list of provinces uses the 2017 data)[13] i.e. North Sulawesi, Southeast Sulawesi, West Sulawesi, Gorontalo, Maluku, North Maluku, West Papua, and Papua are scarce. This suggests the need to increase explorative studies on the living experiences of LAP in population most at risks. This information would be needed to increase a more contextual awareness for the larger public, practitioners, and other researchers. Stigma knowledge in the domains of education and public entitlements in Indonesia are needed. Views and experiences from children and key persons in the structural field both private and government-owned are also essential. In addition, the quality of qualitative literature published in the Indonesian journals also needs to be improved.

Limitations

There are some limitations in this study. Firstly, before it was conducted, the systematic review was not registered on the relevant platform. Even though, a search on registered reviews had been carried out to prevent duplication. Secondly, some articles were obtained from local peer-reviewed journals that were not yet indexed in international databases. Grey literature, that is known to lack peer-reviewing process such as thesis report, were also involved. The authors did not filter literature with these limits in the searching process in order to widen the opportunity in gaining more knowledge to meet the research objective. This was also done to get more Indonesian language-written literature that were so difficult to find in international databases. To make sure all the literature reviewed still meet the minimum quality, authors performed quality checking with CASP with designated standard. Next, only online databases and open-access literature were used in the searching which allowed research reports stored outside of this to be excluded. Finally, although the risk of bias in the interpretation of meanings had been minimized by the researchers who were all Indonesians and with a consensus step to resolve discrepancies, it is worth to note that in a text analysis of qualitative studies, differences in meaning might still happen with the original author of the article.

Conclusion

LAP in Indonesia felt (internalized and anticipated) and experienced stigma in various life domains both in private and public domains. Stigma against LAP was occurred in the intimate relationship, domestic, neighbourhood, up to public arena such as in the occupational, formal education, health care, and other public settings essentials for living. Stigma in the community domain and the experienced type of stigma were predominantly found in the qualitative research conducted in Indonesia. Ten categories of stigma drivers and five areas of stigma consequences were found. The focus of anti-stigma program should make sure that all domains have been touched in the program. The approaches to help LAPs cope with stigma could be adapted to the uniqueness of each domain, including what's there that generate stigma specific to the local culture. Further research is needed to capture the stigma felt and experienced by LAP, its determinants, and consequences outside Java provinces and in education and public entitlements. Another study could also seek if stigma experienced in one domain will lead to the feeling of being stigmatized in another domain of life and if there is an association between the stigma consequences to the occurrence of other stigma.

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Figure Legend:

Figure 1. PRISMA flow diagram depicting the study selection process

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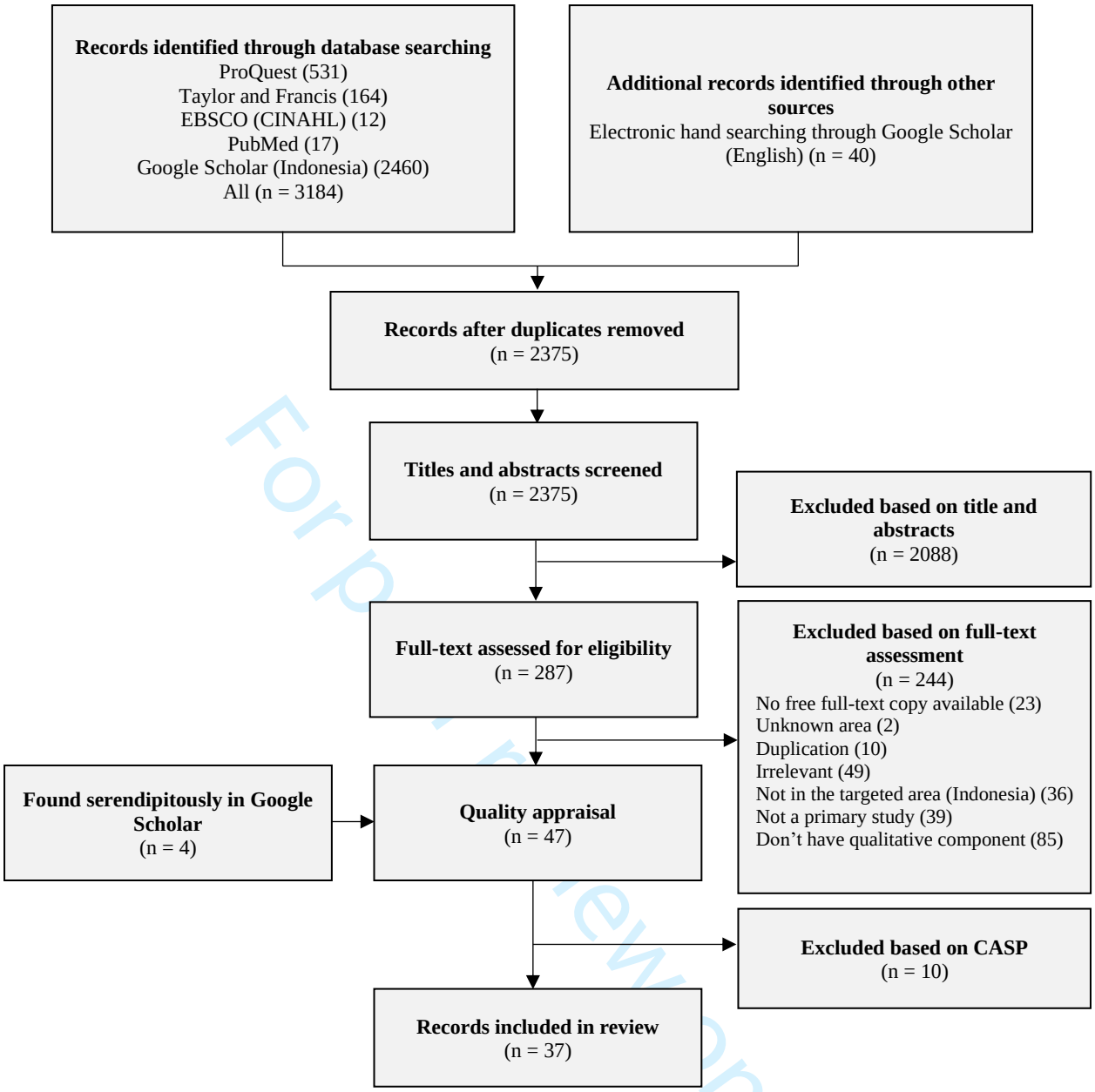


Figure 1. PRISMA flow diagram depicting the study selection process

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Supplement Material 1

Syntaxes for search

Databases	Syntaxes	Results
ProQuest Database	(("risk factor" OR "risk factors") OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR "quality of life" OR manifestation* OR outcome* OR satisfaction) AND (leprosy OR "Hansen's Disease" OR "Hansen Disease" OR leprae) AND (qualitative OR anthropolog* OR cultur* OR "focus group" OR "focus-group" OR ethnography OR "grounded theory" OR interview) AND Indonesia)	531 (Scholarly Journal Article) + 18 (Dissertation /Theses)
Taylor and Francis	(“risk factor*” OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR “quality of life” OR manifestation* OR outcome*) AND (leprosy OR “Hansen's Disease” OR “Hansen Disease” OR leprae) AND Indonesia AND (qualitative OR anthropolog* OR cultur* OR “focus group” OR “focus-group” OR ethnography OR “grounded theory” OR interview)	164 (Scholarly Journal Article)
EBSCO (CINAHL)	(“risk factor*” OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR “quality of life” OR manifestation* OR outcome*) AND (leprosy OR “Hansen's Disease” OR “Hansen Disease” OR leprae) AND Indonesia AND (qualitative OR anthropolog* OR cultur* OR “focus group” OR “focus-group” OR ethnography OR “grounded theory” OR interview)	12 (Scholarly Journal Article)
PubMed	#1“risk factor*”[tw] OR cause*[tw] OR determinant*[tw] OR stressor*[tw] OR belief*[tw] OR construct*[tw] OR practice*[tw] OR driv*[tw] OR facilitat*[tw] #2 impact*[tw] OR effect*[tw] OR influence*[tw] OR result*[tw] OR stigma[tw] OR “quality of life”[tw] OR manifestation*[tw] OR outcome*[tw] OR satisfaction[tw] #4 leprosy[tw] OR “Hansen's Disease”[tw] OR “Hansen Disease”[tw] OR leprae[tw] #5 Indonesia[tw] #6 qualitative[tw] OR anthropolog*[tw] OR cultur*[tw] OR “focus group”[tw] OR “focus-group”[tw] OR ethnography[tw] OR “grounded theory”[tw] OR interview[tw]	17 (Scholarly Journal Article)
Google Scholar (English)	leprosy AND qualitative AND (stigma OR "quality of life") AND ("east java" OR "west java" OR gorontalo OR sulawesi OR papua OR maluku)	40 (Electronic Hand Searching)
Google Scholar (Indonesia)	*kusta AND kualitatif ("faktor risiko" OR penyebab OR determinan OR "sumber daya" OR dampak OR pengaruh OR stigma OR "kualitas hidup" OR sulawesi OR gorontalo OR maluku OR papua OR jawa OR indonesia)*	2460 (All type of publication)

Supplement Material 2

Critical Appraisal Skills Programme (CASP) score for the reviewed studies

No	Study (first author, year)	CASP										
		Was there a clear statement of the aims of the research?	Is qualitative methodology appropriate?	Was the research design appropriate to address the aims of the research?	Was the recruitment strategy appropriate to the aims of the research?	Was the data collected in a way that addressed the research issue?	Has the relationship between researcher and participants been adequately considered?	Have ethical issues been taken into consideration?	Was the data analysis sufficiently rigorous?	Is there a clear statement of findings?	How valuable is the research?	CASP Score
1	Al Bana, 2014[33]	1	1	1	1	1	0	1	1	1	1	9
2	Arisal, 2020[37]	1	1	1	1	Can't tell	1	0	1	1	1	8
3	Armajin, 2019[39]	1	1	1	1	Can't tell	0	0	Can't tell	1	1	6
4	Aulia, 2019[55]	1	1	1	1	Can't tell	1	0	0	1	1	7
5	Dadun, 2016[29]	1	1	1	1	1	0	1	1	1	1	9
6	Dary, 2017[22]	1	1	1	1	Can't tell	0	1	1	1	1	8
7	Fajar, 2002[47]	1	1	Can't tell	1	1	0	0	Can't tell	1	1	6
8	Hidayat, 2020[49]	1	1	1	1	1	0	1	1	1	Can't tell	8
9	Huda, 2011[48]	1	1	1	1	1	0	0	1	1	1	8
10	Indow, 2019[52]	1	1	1	1	1	0	1	1	1	1	9
11	Jatimi, 2020[44]	1	1	1	1	1	0	1	1	Can't tell	1	8
12	Jufriyanto, 2020[23]	1	1	1	1	1	0	1	1	1	1	9
13	Kusharnanto, 2013[35]	1	1	1	1	1	Can't tell	1	1	1	1	9
14	Lusli, 2015[2]	1	1	1	1	1	Can't tell	1	1	1	1	9
15	Lusli, 2016[40]	1	1	1	1	1	1	1	1	1	1	10
16	Lusli, 2017[45]	1	1	1	1	1	1	Can't tell	1	1	1	9
17	Marbaits, 2012[38]	1	1	1	1	1	0	0	1	1	1	8
18	Nainggolan, 2017[32]	1	1	1	1	1	0	0	Can't tell	1	1	7
19	Najmuddin, 2013[53]	1	1	1	1	1	0	1	1	1	1	9
20	Peters, 2013[8]	1	1	1	1	1	Can't tell	1	1	1	1	9
21	Peters, 2014[7]	1	1	1	1	1	1	1	1	1	1	10
22	Peters, 2016[42]	1	1	1	0	1	0	0	Can't tell	1	1	6
23	Pribadi, 2018[41]	1	1	1	1	Can't tell	0	Can't tell	1	1	Can't tell	6
24	Rai , 2020[50]	1	1	1	1	1	1	1	1	1	1	10
25	Rai , 2020[31]	1	1	1	1	1	1	1	1	1	1	10
26	Romadhon, 2020[30]	1	1	1	1	1	1	1	1	1	1	10
27	Saing, 2016[43]	1	1	1	Can't tell	Can't tell	0	0	1	1	1	6
28	Sandi, 2018[27]	1	1	1	1	1	Can't tell	1	1	1	1	9
29	Schuller, 2010[46]	1	1	1	1	1	Can't tell	Can't tell	1	1	1	8
30	Siregar, 2018[26]	1	1	1	1	Can't tell	0	1	1	1	1	8
31	Sodik, 2016[34]	1	1	1	1	0	0	0	0	1	1	6
32	Susanto, 2017[54]	1	1	1	1	1	0	1	1	1	1	9
33	Tarigan, 2019[51]	1	1	1	1	1	0	Cant tell	Cant tell	1	1	7
34	Verayanti, 2016[25]	1	1	1	1	1	0	0	1	1	1	8
35	Wahyuni, 2017[28]	1	1	1	Can't tell	1	0	0	1	1	1	7
36	Widodo, 2018[36]	1	1	1	1	1	0	0	0	1	1	7
37	Yudanagara, 2020[24]	1	1	1	1	1	0	Can't tell	1	1	1	8

Supplement Material 3

Description of stigma manifestations towards LAPs in Indonesia

Enacted
Avoidance[2,8, 22–27,30,31,33,39–42,49,52,53]
Rejection in greetings, handshake or physical touch[8,24,27–29,31,34,52]
Reluctance to talk[26,36,49]
Absences of invitations to gatherings[28,34,36,46]
Restriction to participate[2]
Negative comment, gossip, or mock[25,26,28,29,35,39,40,50,51,54,55]
Spitting on property and nose was covered[24]
Covering nose[24]
Starring[26]
Rejection to live in the same neighbourhood[30]
Expulsion[47,49]
Unfair treatment due to social background[37,50]
Recitation of a specific prayer to reject transmission[48]
Rejection to marry or to build romantic partner[27,28,32,33,39,46,48]
Acceptance and giving suggestion to divorce[40,46]
Prohibition of touching family member[45]
Concealment of diagnosis by family members[31,47]
The view as a family disgrace[25]
Family rejection even though has been cured[32]
Command not to take a bath[27]
Utensils, clothes, bed separation[47]
Suggestion to leave the house[22]
Shortage of job[29,36,39,51]
Rejection from job[2]
Alienation at work[26,40]
Suspension to work[41]
Dismissal from work[8,25]
Refusal to be in the same body of water (e.g. river, water at rice field) [8,28]
Rejection or hesitation to buy items sold by LAP[8,28,42,45,51]
Hesitation, rejection to help, or delayed treatment from healthcare provider[27,29,55]
Hurtful comments at school[8,35,45,51]
Physical distance[31,39,40,47,49]
Absence of transportation at the residential area where LAP lived[28]
Rejection at the market[33]
Rejection from public transportation driver and passengers[24,39,47]
Anticipated
Fear of avoidance[7,8,26,38–40,42,50,51]
Fear of insults, humiliation, or negative comments[2,7,8,26,38,46]
Fear people would reject to meet[7,28,40]
Fear of infecting others[22,29]
Worry people will notice or suspect something wrong[7,39,40,47]
Fear if interactions will make people will feel offended[24]
Worry the deceased body of LAP would not be buried at the public cemetery[33]
Fear family members will afraid of the disease[52]
Fear to be excluded by the family and neighbours[31,39,52]
Worry of being rejected to take care of family[7,40]
Fear romantic partner will leave[29]
Fear difficulty to have a romantic partner[29]
Internalized
Feeling ashamed of self[8,25–28,31,35,40,44,54,55]
Seeing self as an abnormal or different[2,28,44,46]
Seeing self as dirty, ugly, disgusting, unclean[24,28,29,33,40]
Seeing self as an eternal-patient, will not be healed or normal[2,8,24,33,38,45,50,53]
Seeing self as a danger for other[40]
Feeling no one want to marry[28]
Seeing self a poor person and fit to be a beggar[53]
Seeing self as a family troublemaker or burden[44–46]
Seeing self as an object of charity[2,28,46]
Seeing self as having a dangerous disease[53]
Seeing a stamp of ‘leprosy’ at self[24]
Feeling inferior[8,37]
Feeling people will not be friendly anymore[8]
Feeling as a cursed person even after cured[32]
Feeling unworthy[32]
Feeling will not be able to get a job or work again[24,35]
Feeling people are afraid of self[24]

Supplement Material 4

Description of drivers of stigma towards LAPs in Indonesia

Drivers	Description	
Physical factors[2,8,22–25,27,28,31–33,35,39–44,46,49–53,55]	Physical appearance due to deformity or a darkening skin, unpleasant physical odour, decreased physical function, decreased performance due to physical weakening, continuous pain after cured	Related with the disease progress
Fear of contagion[2,7,8,22–24,26–28,31, 34,35,39,43,45,47,52,53,55]	Fear of fast transmission from breathing the same air, standing in the similar body of water, shaking hands, eating food prepared by the person affected, carrying the deceased body of a person, using the same personal object such as glasses, towels, clothes	
Hereditary factor[2,38,46,48,49,53]	Believed that leprosy can be inherited in the family line	
Local connotation for LAP[25,46,47,53]	Negative connotation attached to the local word directed to leprosy affected persons (e.g *'Nibamparekangji'* in South Sulawesi that has a meaning 'to ignore or treat them like they do not exist or *'Judam'* in East Java that means 'ugly and hated by people')	
Religious and/or mystical reason[22,24,31,32,34,45–49,50,53,55]	Believed that leprosy is caused by sin, curse, punishment on bad deeds, evil spirit influence, or spell	
Incapability performing social role[37,40,44–46,51,53]	Lack or no opportunity to perform previous routine or responsibility in and for the family, continuous failure in relationship, job, school	
Factors related to social and health assistance[8,32,38,40,47,55]	Home visits from health officer*[can be considered positive approach, yet could drive stigma]*, exhaustion in caring for patients, health worker communication style, or an inadequate social rehabilitation	
Beliefs around prognosis[36,38,43,45,48,53]	Believed leprosy as a long-term disease, a high fatality disease, and an incurable disease	
Residential area[28,30]	Believed that the residential area where persons-affected live is a dangerous zone or by living there you will also get stigmatized	
Social background[37,45,50]	Ethnic, religion, gender, or economic background	Not related with the disease progress

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Supplement Material 5

Description of stigma consequences on LAPs in Indonesia

Social behaviour
Self-isolation[7,8,22,26–28,30,31,45,47,48–51]
Avoiding new people, home visits, and social activities[2,8,24,27,34,39,40]
Apathy to others[38]
Hiding activities from family members[42]
Difficulty to marry, build romantic relationship or friendship[2,24,32,33,46]
Estranged or damaged relationship with significant others[8,29,32,45]
Being passive at home[2,27,38,40]
Changing place to live[22,24,25,32,35,37,51]
Hiding face or body parts when going outside[28,35,39,40]
Loss of social image[31]
Limited inter-island movement[30]
Being used for political and economic purpose[30,37]
Psychological
Sadness[8,25–28,36,38,45,49,24,51,52]
Frustration[8,24,27,32,35,38,40,51,52]
Despair[8,27,40]
Guilt[44,46]
Felling lonely[2,22,40,45]
Feeling nsafe[39]
Anger or easy to be irritated[32,41]
Inability to be angry[28]
Pessimism[8,25,38,40,50]
Loss of self-confidence[27,34,40]
Feeling unfulfilled[32]
Feeling useless[28]
Being ashamed of self (for a long time period)[8,24,36,37,42,44,51]
Self-disappointment[28,40,51]
Spiritual distress (e.g. questioning God or stopping routine religious activity)[27,32]
Loss of passion to live[29,32,38,42]
Suicidal thinking[7,8,22,24,28,33,38–40]
Suicidal attempt[38]
Occupational/financial
Resigning from work[8,26,40]
Difficulty to find job[2,29,35,36]
Losing customer[8,42,51]
Preference to work at night[24]
Feeling shy to find a proper job[25,35]
Weak financial condition[29,37,45]
Dependence on aid to sustain life (e.g. becoming a beggar or leaning on government's aid)[25,29,37,53]
Health
Concealing, lying or denying the diagnoses[2,7,8,26,30,33,36,37,48,49,52]
Avoiding healthcare facility or provider[8,28,30,31,38,39,47,53,55]
Abandoning the sign of leprosy[38,47]
Seeking treatment secretly[47]
Seeking traditional healer[31,47]
Medication noncompliance[31,47]
Lack of family support in treatment[47]
Absence in health education[53]
Unhygienic practice of self-care[27]
Worsening disease[31,41,55]
Weakened physical health[38,41]
Excessive use of drugs[51]
Neglecting general health[53]
Self-mutilation[7]
Education
Stop attending school[8,28,29,35,40,45,51]
Difficulty to access school bus[28]
Having teachers who refuses to live on the same island with the students[30]

Supplement Material 6

Coverage of domain impacted with stigma in the included literature

Themes	Literature																																							
Affected private domains	**Number of studies**	33	37	39	55	29	22	47	49	48	52	44	23	35	2	40	45	38	32	53	8	7	42	41	50	31	30	43	27	46	26	34	54	51	25	28	36	24
Community	35/37	•	•	•	–	•	•	•	•	•	•	•	•	•	•	•	–	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Domestic	22/37	•	•	•	–	–	•	•	–	–	•	•	•	–	•	•	•	–	•	–	•	•	–	–	–	•	–	–	•	•	•	–	–	•	•	•	–	•
Intimate relationship	13/37	•	•	•	–	•	•	•	–	•	–	–	–	–	–	•	•	–	•	–	–	–	–	–	–	–	–	–	•	•	–	–	–	–	–	•	–	–
Affected public domains																																						
Healthcare	15/37	–	–	–	•	•	–	–	–	–	•	•	–	•	–	–	–	•	•	•	•	–	–	–	–	•	•	–	•	-	•	–	•	•	–	•	–	–
Economic	20/37	–	•	•	–	•	–	–	–	–	–	–	–	•	•	•	•	–	•	•	•	•	•	•	–	–	•	–	•	-	•	–	–	•	•	–	•	•
Education	8/37	•	-	•	–	–	–	–	–	–	–	–	–	•	–	•	•	–	–	–	•	–	–	–	–	–	–	–	–	–	–	–	–	•	–	•	–	–
Public entitlements	6/37	•	–	•	–	–	–	•	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	•	–	–	–	–	–	–	–	–	–	–	–	•	–	•

• : mentioned in the literature

-: not mentioned in the literature

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2 & attached document "Abstract Checklist"
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3 (paragraph 1-3)
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 3 (paragraph 3)
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 3
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 3 (search strategy)
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 4 (search strategy)
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4 (study selection)
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 4 (data extraction and analysis)
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 4 (data extraction and analysis)
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 4 (data extraction and analysis)
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4 (study selection), Supplement material 2 (appraisal)
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	N/A
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 4 (study selection)
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 5 (synthesis)
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 5 (synthesis)
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 5 (data extraction and analysis)
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 4 (search results)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Page 4,5,and table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplement material 2 (appraisal)
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	N/A
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Supplement material 2 (appraisal)
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 12-13
	23b	Discuss any limitations of the evidence included in the review.	Page 13 (limitation)
	23c	Discuss any limitations of the review processes used.	Page 13 (limitation)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	23d	Discuss implications of the results for practice, policy, and future research.	Page 13 (paragraph 6)
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 13 (limitation)
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 14 (funding and acknowledgement)
Competing interests	26	Declare any competing interests of review authors.	Page 14
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 14 (data sharing statement)

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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TITLE PAGE

Stigma towards leprosy across seven life domains in Indonesia: a qualitative systematic review

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Abstract

Objective: To provide knowledge about the domains of life affected by stigma towards leprosy in Indonesia, including its manifestations, driving factors and consequences.
Design: Qualitative systematic review.
Study selection: PubMed, CINAHL, ProQuest, Taylor&Francis, and Google Scholar were used to systematically search studies with qualitative component that were conducted in Indonesia and published from January 2000 to December 2020 in English or Indonesian language. The search was started in November 2020 and reran in April 2021. Quality assessment and thematic synthesis were applied.
Data extraction: Of the 3184 studies, 37 manuscripts were reviewed. Information relating to study characteristics, stigma domains and types following Weiss Extended Scambler's Hidden Stress Model, stigma consequences, and drivers were extracted.
Results: Seven themes were identified. Three—community, domestic, and intimate relationships—impacted private domains. Four— health, economics, education, and public entitlements—concerned public domains. Studies mainly discussed enacted stigma rather than anticipated and internalized stigma. Ten stigma-driving factors were found, ranging from negative and positive concepts linked with the condition to aspects not related to the disease process. Five areas of consequences were shown. Impact on public rights, such as education, were very minimally explored, although school absence was often mentioned. Stigma manifestation, drivers, and consequences in most public domains were least explored.
Conclusion: Leprosy-affected persons in Indonesia experienced and felt stigma in private and public domains. Disease-related aspects, the culture and history of a particular region are linked with stigma manifestations. Approaches in one domain can affect another domain. More exploratory studies are needed in the endemic areas outside Java, especially considering both the lack of studies there and the unique culture of each Indonesian region.

Keywords
domain of life, leprosy, stigma, Indonesia, qualitative, review

Strength and limitations of this study

- Studies written in both English and Indonesian are included to provide richer information about the Indonesian context and to complement previous reviews.
- Findings were derived from various sources including scholarly journal articles and academic theses.
- In textual analysis of qualitative studies, differences in meaning with the original author of the article are possible, although this risk was minimized by the fact that all researchers were Indonesian and because an attempt to reconcile discrepancies was made.

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Introduction

Though a treatment for leprosy has existed for decades,[1] life as a leprosy-affected person (LAP) has never been easy.[2] A strong stigma is attached to leprosy. Studies have shown that the life burden of affected persons is greatly influenced by stigma rather than the disease itself. Stigma affects not only people infected, but also those affiliated with them.[3] This is due to perceptions of leprosy as a contagious disease, a consequence for violation of moral or divine instruction[4] and a debilitating disease marked by paralysis.[5] These concepts still exist. Therefore, LAP's experiences of rejection, exclusion, and the inability to disclose their condition can be found in recent literature,[4, 6–10] including in Indonesia.[9, 10]

In Indonesia, leprosy has a long history and remains a burden. For centuries, leprosy has affected all five of Indonesia's big islands.[7] Although significant improvements in leprosy treatment in Indonesia have caused a gradual decline in cases, the decline in the number of LAPs has been slow over the last decade.[11] According to the latest report, Indonesia has the third-highest number of leprosy cases globally, after India and Brazil.[12] 10 out of 32 Indonesian provinces, and 142 districts/cities, have not yet reached elimination.[11, 13] One cause for this is stigma and discrimination.[13]

Knowledge of stigma towards LAPs in the context of Indonesia is needed. Jones *et al.* and Yang *et al.* emphasize that stigma can be understood through examining norms surrounding what is disliked and unwanted in the local social setting.[14, 15] Indonesia is culturally rich, as reflected in its 300 ethnic groups spread across 17,504 islands and 32 provinces.[16] Because the spread of leprosy is not centralised in a specific part of the country, there is likely variation pertaining to leprosy-related stigma throughout the country. Reviews on this topic were difficult to find, particularly in the international literature.[9] In addition, recent reviews in the area of stigma and leprosy did not include Indonesian-language studies, and appeared to generalize findings from different contexts, mainly from English-language literature.[17, 18] This review aims to fill that knowledge gap. The study identified, synthesized, and discussed qualitative literature in the field of leprosy in Indonesia in order to provide knowledge about the domains of life affected by stigma towards leprosy-affected people in Indonesia, including stigma manifestations in each domain, driving factors, and consequences.

Method

Literature search

Eligibility criteria

Inclusion criteria were qualitative studies and mixed-method studies with a qualitative component conducted in Indonesia and published in Indonesian and English. Only studies using primary data, and in which a full-text online copy was available, were incorporated. Publication types included were peer-reviewed scholarly articles from national and international journals, and grey literature (i.e. academic theses or dissertations and manuscripts residing in university repositories in Indonesia), published within the past 20 years (2000-2020).

Search strategy

This search was guided by a PICo-formulated research question: "What are the leprosy-affected persons' (Population) domains of life affected by stigma, including stigma manifestations, driving factors and consequences (Interest) in Indonesia (Context)?" This review complies with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[19] Studies were searched in PubMed, CINAHL, Taylor and Francis, ProQuest Databases, and Google Scholar (see supplement material 1. Syntaxes for search). The use of Google Scholar complemented literature findings, since many local studies, written in Indonesian, were absent from other databases. Google Scholar is also connected with a large amount of grey literature from local universities' repositories and local peer-reviewed journals.

Study selection

Firstly, titles and abstracts of retrieved literature were screened to remove duplicates. Studies not in line with eligibility criteria were removed. Next, study quality was appraised using the Critical Appraisal Skills Programme (CASP) tool for qualitative research.[20] CASP consists of ten questions which discusses three

broad issues in the qualitative study: results' validity, quality, and usefulness. Each question comprises three answers, for which researchers used a scoring system: 1 for "yes", and 0 for "no" or "can't tell". In the table presentation (see supplement 2. Critical Appraisal Skills Programme (CASP) score for the reviewed studies), "can't tell" is distinguished from "no." Studies not addressing at least six criteria of CASP were not included in synthesis. Discrepancies between researchers were resolved through discussion.

Data extraction and analysis

Extraction and synthesis of the data were conducted. The following characteristics were extracted: study design and method, study perspective, number of participants, age range, and study location. Extraction also included variables reflecting the review domains: type of stigma (internalized, anticipated, or enacted) following Weiss's (2008) Extended Scambler's Hidden Stress Model, setting of stigma, consequences of stigma, and drivers of stigma.[21] Thematic synthesis was used to analyse these data.[22] The first step was free line-by-line coding of text, guided by aforementioned review domains. Then, organisation of codes under each review domain were conducted. Codes were grouped into descriptive themes. Finally, analytical themes were generated to emphasize experience in each stigma-affected domain by reviewing findings under each setting affected. Literature was read repeatedly to ensure all concepts needed to answer research objective were mapped. Each step was done manually by YMM, then reviewed by ER and ATD. Paper, pen, and Ms. Excel were used in the process. Final descriptive and analytical themes derived from review were discussed and agreed-upon between reviewers (YMM, ER). In addition, a coverage analysis was conducted to search knowledge gaps in the affected life domains.

Patient and Public Involvement

Not applicable, as it was not appropriate or possible to involve patients or the public in the design, conduct, reporting, or dissemination plans for a review study.

Ethical statement

Not applicable. This study is a review of articles.

RESULTS

Search results

The literature finding (Figure 1) resulted in 3184 studies, of which 2375 were left following removal of duplicates. From screening of titles and abstracts, 2088 studies were excluded. A full-text assessment for eligibility was conducted for 287 articles. Remaining articles for CASP appraisal included 47 total studies, including 4 eligible articles from serendipitous findings. After CASP appraisal (see supplement material 2), 37 studies remained for synthesis.

Description of the included studies

The majority of the 37 studies (Table 1), had been published in the past five years, from 2015 to 2020 (n=28). Most were qualitative (n= 34). Three were mixed-methods. Only a few (n=17) clearly stated the design used. Phenomenology was used most often (n=9). Others were case studies (n=6), comparative ethnographies, (n=1) and life history (n=1) respectively. The majority took place in Java Island (n=29). A limited number of conducted from provinces outside Java i.e., South Sulawesi (n=5) with one study each from North Maluku, North Sumatera, and West Papua provinces. Some studies did not provide detailed information about frequency of data collection, yet at least 560 interviews and 27 focus groups were identified from the reviewed literature. 21 studies were published in Indonesian and 16 in English. 26 were journal articles and 11 studies were academic theses.

Synthesis

Domains of life penetrated by stigma manifestations, drivers, and consequences towards LAP in Indonesia were synthesized (Table 2). The seven impacted domains were 'community', 'domestic', 'intimate relationship', 'health', 'economic', 'education', and 'public entitlements'. Seven themes were developed to explain stigma manifestations, drivers and consequences in each domain. The description of stigma manifestations (see supplement material 3. Description of stigma manifestations towards LAPs in Indonesia), ten categories of stigma drivers on LAP in Indonesia (see supplement material 4. Description

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stigma drivers towards LAPs in Indonesia) and five areas of stigma consequences were also elaborated (see supplement material 5. Description of stigma consequences on LAPs in Indonesia). Next, findings were explained narratively to enable comprehension of the effect of stigma in each domain of the LAP's life. In each domain, stigma manifestation and driving factors were addressed first, followed by consequences.

Stigma in private domains

Community: difficult relationship and alterations in social standing

LAPs in Indonesia face many difficulties in building a satisfactory relationship with their communities, indicating an effect on their social standing. Issues were experienced with friends,[23, 24] neighbours,[9, 25–27] or others in the hometown.[25] These issues were not limited to adults.[28, 29] Enacted,[2, 9, 10, 24–26, 28–47] anticipated,[2, 9, 10, 25, 27–29, 36–38, 45, 48–51] and internalized stigma,[2, 9, 10, 26, 38, 41, 45, 48–50, 52, 53] were linked to this.

Stigma manifestation and driving factors at the community level

Forms of enacted community stigma were avoidance meeting,[2, 29, 30, 34, 38, 48, 54] visiting or hosting,[9, 27, 44, 49] greeting,[28] talking with,[45, 47, 50, 54] being physically near,[24, 30, 35, 40, 50] having contact with,[9, 44, 45, 53] and even looking at LAPs.[27] Rejection and hesitation toward LAPs were experienced with an un-returned smile and hand offer,[2, 9, 28] an insincere handshake gesture,[27] a recitation of prayer over the food made by the LAP,[49] an unwillingness to live in the same neighbourhood,[25, 37, 41, 49] and to be immersed in the same body of water.[29] Exclusions were mentioned.[23, 28, 29, 35, 44, 46, 48, 54] These included restrictions on community participation[2] and an absence of invitations to gatherings.[29, 43, 45] LAPs also experienced expulsion,[50] suggestions to leave the community,[25] gossiping,[29, 31, 34] verbal mockery,[23, 28, 29, 31, 33, 40, 48, 52, 55] and non-verbal insults such as people covering their noses and spitting when passing the LAP's house.[27]

Drivers of these enacted stigma were associated with physical factors,[2, 23, 26, 29, 30, 40, 42, 50–53] fear of contagion,[26, 28, 45, 49, 53] religious or mystical reasons,[27, 40, 44, 49] local connotations of the word 'leprosy' labelling the disease as fatal, ugly, insulting, or worthy of ignoring,[43, 48] inhabiting the residential area known for leprosy,[41] health worker home visits,[9, 35] and inadequate social rehabilitation following release from treatment.[37] Conditions unrelated to leprosy, such as a poor financial conditions,[51] being a woman, or belonging to an ethnic or religious minority were also linked to unfair treatment.[46]

The anticipated stigma among LAPs in the community domain were fear of avoidance,[2, 9, 10, 36, 40, 47] rejection,[29, 34, 47] exclusion,[40, 47] insult,[2, 9, 10] rumours,[10] and irritating others by initiating conversation.[27] Limited studies mentioned reasons for this anticipated stigma, i.e. changes in physical feature,[29, 35] heredity,[54] beliefs that leprosy is contagious, a curse, a spell, a trial, or a sin,[54] and negative past experiences in the community.[27]

Internalized stigma include self-labelling, self-deprecation, and shame. LAPs perceived themselves as disgusting,[29] dirty,[35] dangerous,[35] contagious,[29] objects of charity,[43] incurable,[9] different,[32] and inferior.[46] They perceived that they could not expect friendliness,[9] and mentioned being an enemy of the community.[49] LAPs in this stage could feel others' fear of them,[27] felt awkward gathering with others,[52] and viewed self-retraction from the community as reasonable.[10] Physical appearance was the most-repeated reason for internalized stigma.[23, 35, 39] Others included beliefs about disease transmission[10] and witnessing a healthcare worker's (HCW) avoidance.[9]

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Table 1 Description of studies included in the review

Research (first author, year)	Province(s)	City/district(s)	Study design, type	Data collection method(s)	Study perspective(s)	Age criteria of LAP included in the research
Al Bana, 2014[30]	East Java	Mojokerto	Life history, qualitative	Interviews	LAP	Adult
Arisal, 2020[46]	South Sulawesi	Makassar	Design not stated, qualitative	Interviews, observation and documentation	LAP, family, and community	Adolescent
Armaijn, 2019[47]	North Maluku	Ternate	Design not stated, qualitative	Interviews	LAP	Adult
Aulia, 2019[56]	East Java	Mojokerto	Design not stated, qualitative	Interviews and observation	LAP and health worker	Age not clearly addressed, possibly adult
Dadun, 2016[31]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family, community, religious leader, health worker, social and business worker	Age not clearly addressed, possibly adult
Dary, 2017[25]	Central Java	Jepara	Phenomenology, qualitative	Interviews	LAP, health worker	Adult, adolescent
Fajar, 2002[48]	East Java	Gresik	Design not stated, mixed-methods	Interviews	LAP	Adult
Hidayat, 2020[50]	East Java	Sumenep	Phenomenology, qualitative	Interviews	Community	Adult
Huda, 2011[49]	East Java	Sampang	Design not stated, qualitative	Interviews and observation	LAP, community, health worker	Adult
Indow, 2019[53]	West Papua	Manokwari	Case study, qualitative	Interviews	LAP	Adult
Jatimi, 2020[32]	East Java	Sumenep	Phenomenology, qualitative	Interviews	LAP	Adult
Jufriyanto, 2020[26]	East Java	Pamekasan	Phenomenology, qualitative	Interviews	LAP	Adult
Kusharnanto, 2013[33]	Central Java	Blora	Design not stated, qualitative	Interviews, observation and documentation	LAP	Adult
Lusli, 2015[2]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus groups	LAP	Age not clearly addressed, possibly adult
Lusli, 2016[35]	West Java	Cirebon	Design not stated, mixed-methods	Interviews at baseline and post	LAP	Age not clearly addressed, possibly adult
Lusli, 2017[34]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family (caregiver), counsellors	Age not clearly addressed, possibly adult
Marbaits, 2012[36]	East Java	Gresik	Case study, qualitative	Interviews and observation	LAP and family	Adult
Nainggolan, 2017[37]	Banten	Tangerang	Case study, qualitative	Interviews and documentation	LAP, community, health worker, and social worker	Adult
Najmuddin, 2013[54]	South Sulawesi	Makassar	Phenomenology, qualitative	Interviews	LAP	Age not clearly addressed, possibly adult
Peters, 2013[9]	West Java	Cirebon	Design not stated, qualitative	Interviews and focus group	LAP, family, community, religious leader, health worker	Adult, child
Peters, 2014[10]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adult
Peters, 2016[38]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adult
Pribadi, 2018[39]	East Java	Pasuruan	Case study, qualitative	Interviews, observation and documentation	LAP, family, and community	Adult
Rai , 2020[51]	West Java	Cirebon	Design not stated, mixed-methods	Interviews	LAP	Adolescent, adult
Rai , 2020[40]	West Java	Cirebon	Design not stated, qualitative	Interviews	LAP	Adolescent, adult
Romadhon, 2020[41]	East Java	Sampang and Mojokerto	Comparative ethnographic, qualitative	Interviews and observation	LAP, family, community, health worker	Age not clearly addressed, possibly adult
Saing, 2016[42]	South Sulawesi	Palopo	Design not stated, qualitative	Interviews and observation	LAP, community, health worker, government worker	Age not clearly addressed, possibly adult
Sandi, 2018[28]	East Java	Ngawi	Phenomenology, qualitative	Interviews	LAP	Adult
Schuller, 2010[43]	South Sulawesi	Gowa and Maros	Design not stated, qualitative	Interviews and focus group	LAP, community, and religious leader	Adult
Siregar, 2018[24]	West Java	Depok	Phenomenology, qualitative	Interviews and observation	Family	Adult
Sodik, 2016[44]	East Java	Trenggalek	Design not stated, qualitative	Interviews and observation	LAP, family, community, organization (co-worker)	Adult
Susanto, 2017[55]	East Java	Jember	Phenomenology, qualitative	Interviews	LAP	Adult
Tarigan, 2019[52]	North Sumatera	Karo	Design not stated, qualitative	Interviews, observation and documentation	LAP	Adult
Verayanti, 2016[23]	South Sulawesi	Makassar	Case study, qualitative	Interviews, observation and documentation	LAP, family, community, health worker	Adolescent, adult
Wahyuni, 2017[29]	East Java	Sumberglagah	Case study, qualitative	Interviews	LAP	Adult
Widodo, 2018[45]	East Java	Nganjuk	Design not stated, qualitative	Interviews, observation and documentation	LAP, community, health worker	Adult
Yudanagara, 2020[27]	Central Java	Jepara	Phenomenology, qualitative	Interviews	LAP	Adult

Table 2 Domains, manifestations, consequences, and drivers of stigma towards LAPs in Indonesia

Research (first author, year)	Domain(s) where stigma impacts*	Stigma manifestation(s)	Stigma consequence(s) on LAPs	Stigma driver(s)
Al Bana, 2014[30]	C, D, IR, H, PE	Enacted, Anticipated, Internalized	Social behaviour, Psychological	Fear of contagion, Physical factors
Arisal, 2020[46]	C, D, IR, EC	Enacted, Internalized	Social behaviour, Psychological, Occupational/Financial	Social background, Inabilioty to perform gender role
Armaijn, 2019[47]	C, D, IR, EC, ED, PE	Enacted, Anticipated, Internalized	Social behavioural, Psychological, Health	Physical factors, Fear of contagion
Aulia, 2019[56]	H	Enacted, Internalized	Psychological, Health	Fear of contagion, Social and health assistance factors, Physical factors, Religious and/or mystical reason
Dadun, 2016[31]	EC, IR, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial	Social background, Fear of contagion, Physical factors, Religious and/or mystical reason
Dary, 2017[25]	C, D, IR	Enacted, Anticipated	Social behaviour, Psychological	Religious and/or mystical reason, Fear of contagion, Physical factors
Fajar, 2002[48]	D, IR, C, ED, PE	Enacted, Anticipated	Health	Local connotation for LAP, Religious and/or mystical reason, Physical factors, Social and health assistance factors
Hidayat, 2020[50]	C	Enacted	No consequence mentioned	Fear of contagion, Physical factors, Hereditary factor, Religious and/or mystical reason
Huda, 2011[49]	C, IR	Enacted, Internalized	No consequence mentioned	Hereditary factor, Beliefs around prognosis, Fear of contagion, Religious and/or mystical reason
Indow, 2019[53]	D, C	Enacted, Anticipated, Internalized	Psychological	Physical factors, Fear of contagion
Jatimi, 2020[32]	D, C	Internalized	Social behaviour, Psychological	Inability to perform social role, Physical factors
Jufriyanto, 2020[26]	C, D	Enacted	No consequence mentioned	Fear of contagion, Physical factors
Kusharnanto, 2013[33]	C, EC ED, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Education	Physical factors, Fear of contagion
Lusli, 2015[2]	C, D EC, H	Enacted, Anticipated, Internalized	Social behaviour, Health, Psychological, Occupational/Financial	Physical factors, Fear of contagion
Lusli, 2016[35]	C, D, IR, EC, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health, Occupational/Financial	Physical factors, Incapability to perform social role, Social and health assistance factors
Lusli, 2017[34]	D, IR, EC, ED	Enacted, Internalized	Social behaviour, Psychological Health, Occupational/Financial, Education	Hereditary factor, Religious and/or mystical reason, Beliefs around prognosis, Inability to perform social role, Fear of contagion, Social background
Marbaits, 2012[36]	C, H	Anticipated, Internalized	Social behaviour, Psychological, Health	Social and health assistance factors, Hereditary factor, Beliefs around prognosis
Nainggolan, 2017[37]	C, D, IR, EC, H	Enacted, Internalized	Social behaviour, Occupational/Financial, Psychological	Religious and/or mystical reason, Physical factors, Social and health assistance factors
Najmuddin, 2013[54]	C, H, EC	Anticipated, Internalized	Psychological, Health, Occupational/Financial	Physical factors, Inability to perform social role, Local connotation for LAP, Hereditary factor, Fear of contagion, Religious and/or mystical reason, Beliefs around prognosis
Peters, 2013[9]	D, C, EC, ED, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological Occupational/Financial, Health, Education	Physical factors, Social and health assistance factors, Fear of contagion
Peters, 2014[10]	D, C, EC	Anticipated, Internalized	Social behaviour, Psychological, Health	Physical factors, Fear of contagion
Peters, 2016[38]	C, D, EC	Enacted, Anticipated	Social behaviour, Psychological, Occupational/Financial	Physical factors
Pribadi, 2018[39]	C, EC, PE	Enacted, Internalized	Social behaviour, Psychological, Health	Physical factors
Rai , 2020[51]	C, H	Enacted, Internalized	Psychological	Religious and/or mystical reason, Physical factors, Social background
Rai , 2020[40]	C, D, H	Enacted, Anticipated, Internalized	Social behaviour, Health	Religious and/or mystical reason, Fear of contagion, Physical factors
Romadhon, 2020[41]	C, EC, H	Enacted, Anticipated, Internalized	Social behaviour, Health	Residential area
Saing, 2016[42]	C	Enacted	No consequence mentioned	Physical factors, Fear of contagion, Beliefs around prognosis
Sandi, 2018[28]	D, C, IR, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health	Physical factors, Fear of contagion
Schuller, 2010[43]	C, D, IR	Enacted, Anticipated, Internalized	Social behaviour, Psychological	Inability to perform social role, Physical factors, Religious and/or mystical reason, Hereditary factor, Local connotation for LAP
Siregar, 2018[24]	D, C, EC, H	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Health	Fear of contagion
Sodik, 2016[44]	C	Enacted	Social behaviour, Psychological	Religious and/or mystical reason, Fear of contagion
Susanto, 2017[55]	C	Enacted	Social behaviour	No drivers mentioned
Tarigan, 2019[52]	C, D, EC, H, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial, Health, Education	Physical factors, Inability to perform social role
Verayanti, 2016[23]	D, C, EC	Enacted, Internalized	Social behaviour, Psychological, Occupational/Financial	Local connation for LAP, Physical factors
Wahyuni, 2017[29]	C, D, IR, EC, H, PE, ED	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Health, Education	Physical factors, Fear of contagion, Residential area
Widodo, 2018[45]	C, EC	Enacted	Social behaviour, Occupational/Financial	Fear of contagion, Beliefs around prognosis
Yudanagara, 2020[27]	C, D, EC, PE	Enacted, Anticipated, Internalized	Social behaviour, Psychological, Occupational/Financial	Religious and/or mystical reason, Physical factors, Fear of contagion

**Community (C):* relationship with neighbours, friends, community members; *Domestic(D):* impacts in household affairs, family roles, relations with family member and relatives; *Intimate relationship (IR):* issues around spouse or romantic relationship; *Health (H)*: healthcare worker approach, ambience in health facility, effect of disfigurement for healthcare; *Economic(EC):* income-related aspects, co-worker relationship; *Education (ED):* responses from school society; *Public entitlements (PE)*: access to and acceptance in public transport, market, kiosk, and worship place.

Stigma consequences in the community level

Stigma in the community domain impacted LAP's mental and physical health, social behaviour, and financial status. LAPs experienced sadness,[2, 23, 27, 28, 35] stress,[27, 36] hurt,[23, 27, 33] self-blame,[35] loneliness,[25, 35] irritation,[39] feelings of being unsafe,[46] inability to imagine a future,[35] loss of passions,[31] and suicidal thoughts and actions.[10, 27, 35, 36 36] These effects could linger at length.[9] At the social behavioural level, LAPs faced difficulty making friends,[2, 27] and practicing local greetings or gestures.[2, 45] They isolated themselves,[2, 9, 23, 25, 28, 29, 35, 36, 39, 40, 55] felt reluctant to share emotional burdens with others,[2] or to become close with new people.[27] They lacked motivation for social activities,[9] became passive at home,[2] reduced activities outside,[9, 28, 51, 52] or if they did venture out, concealed this from family.[38] If LAPs participated in activities outside, they hid deformities by concealing hands under armpits and wearing long pants, long sleeves, hijabs, or hats.[29, 47] Due to stigma, LAP's social standing and reputation declined in the community.[40] LAPs might choose to leave, or have family suggest leaving the neighbourhood and moving to another community in search of acceptance.[23, 33, 37] Stigma in the community domain also endangered physical health. LAPs concealed their disease status,[2, 10, 24, 40, 48, 54] even from family members,[38] denied symptoms,[36, 48] lied about the disease,[24, 47] self-mutilated,[10] rejected HCW's home visits,[35] hesitated or secretly sought treatment,[47, 48] and feared going to the health centre.[48]

Domestic: devaluation of status, roles, and control over own body

Diminished social status of LAPs occurred within the family. This manifested in enacted,[9, 10, 28–30, 32, 46–50, 54, 55] anticipated,[9, 10, 24, 30, 35, 43, 47, 53] and internalized stigma.[2, 29, 32, 34, 43, 47, 52]

Stigma manifestation and driving factors at the domestic level

Studies noted feelings of embarrassment at having LAPs as family members and attempts to conceal their diagnoses.[23, 40, 48] LAPs also received unpleasant treatment from their children.[26, 29, 48] Children hesitated to perform the *'Salim'* tradition (a form of respectful greeting to parents or older people in some regions of Indonesia) to affected parents,[29] requested parent hide when peers visited the house,[29] and stopped attending school because of shame over the affected parents.[48] In the house, LAP experienced negative comments and exclusion from household roles[2, 35, 43] and were not allowed to touch family members.[10, 34, 35, 47] Unwelcoming gestures[48] even after LAPs were declared cured,[27, 37] feelings that family members feared talking with LAPs,[24] avoidance from extended family members,[24, 25] siblings,[28, 30] and in-laws[25] also occurred. In times of illness, an affected elderly felt no family members was willing to visit.[9] Other manifestations were exclusion from family events,[43] expulsion from home,[48] and forms of separation illustrating reduced bodily autonomy such as separate cutlery, beds, clothes[43, 48] and houses,[32, 48] demands that LAPs to clean their belongings,[43, 48] and restricting LAPs from bathing.[28] Enacted stigma in the domestic domain was driven by LAP's physical appearance,[25, 29] long-held local connotations about LAPs,[48] concern over blood transmission,[24] ineffective post-treatment rehabilitation for LAPs and their families,[37, 48] and community beliefs that leprosy is a curse or sin[25] in turn inducing familial fear of reputational damage due to the presence of the LAP.[48]

Anticipated stigma in the domestic domain manifested in fear of avoidance,[24, 47] exclusion,[47] negative comments,[2, 53] and other reactions.[53] Concerns over inability to perform gender roles,[35] restrictions on touching children,[10] and humiliation faced by family members were mentioned.[43] Almost no studies discussed driving factors for anticipated stigma in the domestic domain. One reason may be beliefs that leprosy is an incurable curse.[43]

Devaluation of domestic roles and status were also reflected in internalized stigma. LAPs expressed shame with family members,[29] self-doubt when carry out family responsibilities[34] e.g. breastfeeding, touching children, cooking, or meeting a financial need.[10] The literature also described LAP's seeing themselves as burdens or trouble-makers,[2, 32, 36, 43] irresponsible breadwinners,[52] and sources of transmission to family.[10, 47] Self-stigma was also shown in withdrawal from family and separation of eating utensils and laundry.[47] Fear of transmission,[47] failure to enact gender roles,[29, 32, 43] and changes to appearance[29] were found to promote these behaviours.

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Stigma consequences at the domestic level

Stigma in the domestic domain caused mental, social, and health impacts. Mentally, LAPs experienced sadness,[9, 27, 34, 37, 52, 53] frustration,[27, 37] disappointment,[27, 52] anger,[37] despair,[34] loneliness,[25, 34] hopelessness,[37] guilt,[32, 43] and shame.[52] At the social-behaviour level, 'distance' from family members was felt.[34] LAPs isolated themselves from family,[37] chose not to communicate with them,[2, 24] or moved elsewhere.[23, 27, 37] Stigma at this domain also brought health risks. LAPs concealed and lied about the disease to the family,[35, 47, 53] not routinely taking medicine,[48] while neglecting personal hygiene.[28]

Intimate relationship: trouble in having a life partner

Enacted,[25, 29, 30, 34, 37, 43, 46–49] anticipated,[31, 35] and internalized stigma[29, 43] affected LAP's chances at romantic partnership, marriage, or preservation of existing marriage. Stigma may also impact LAP's family members' ability to have a life partner.[43]

Stigma manifestation and driving factors t the intimate relationship level

In enacted form, studies noted family rejections of LAP's romantic relationships. Disapproval at marriage to an affected person,[37] a non-affected person,[30] or a person who has a family history of leprosy[43, 48, 49] were mentioned. Findings revealed one cancellation of a planned marriage[37, 49] even after an official declaration that leprosy had been 'cured'[37] and one romantic relationship that was ended.[37]. Married LAPs reported avoidance from a spouse,[25, 34] family members' urgings to divorce after diagnoses,[35, 48] and an expulsion from parent-in-law.[48] Reasons for this behaviour ranged from beliefs that leprosy is inherited[43, 49, 50, 54] or a curse,[25] to local connotations of leprosy as ugly or insulting disease,[48] to fear of harming familial reputation,[48, 49] effort to protect the family name,[49] and concern over contagiousness.[25, 30] One study noted that women might face particular problems finding a life partner, due to a stereotype that affected women cannot serve their husband and household optimally.[46]

There is a lack of research discussing anticipated and internalized stigma within this domain. Fears of partners leaving or avoiding LAP's were forms of anticipated stigma[31, 35]. Internalized stigma manifested in feelings of shame around building relationships[43] or in fears of being unwanted in marriage.[29] Physical appearance was reported as the cause for these fears.[29, 43]

Stigma consequences at the intimate relationship level

Issues in intimate relationships resulted in physical, mental, and social risks. Studies mentioned negative emotions such as sadness,[34] stress,[37] despair,[28] feelings of being stuck or unfulfilled,[37] and suicidal thinking.[47] In social life, changes occurred in relationships amongst family members, leading LAP's to leave their households.[37] Risks in health may occur when LAPs hide their disease from a partner.[35]

Stigma in public domain

Health: challenges in healthcare-seeking and disfigurement effect

Stigma in the health domain was demonstrated by the behaviour of people at health facilities and by LAP's responses to disease development. Enacted,[9, 24, 28, 29, 31, 40, 56] anticipated,[41] and internalized stigma[33, 36, 52, 56] were found.

Stigma manifestation and its driving factors in the health level

Enacted stigma was demonstrated among HCWs who refrained from handshaking or physical contact with LAPs,[9, 29, 31, 40] delayed health service,[28, 31] hesitated,[56] or declined to perform treatment,[28, 56] and delivered hurtful words.[56] LAPs experienced staring from visitors at the health facility as well as HCW stigma.[24] Furthermore, prioritizing of men over women in health settings occurred.[46] Physical appearance,[29] concern about transmission,[24, 31, 56] general insensitivity within the health care centre,[56] and a belief in men's greater competence and capability[46] were all cited as driving factors in enacted stigma. The concern of HCW about negative reactions from family members when handling LAP may also explain why their approach was perceived unpleasant by LAP.[31]

An anticipated form of stigma in the health domain was fear of being associated with leprosy if visiting a free health event in the local area.[41] This fear was likely facilitated by a communal desire to reject a

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stereotype surrounding their territory (Mandangin island), where the free health clinic was held. For generations, the area has been known to many as an 'island for leprosy-affected'.[41]

In the health domain, physical appearance effects related to leprosy and its treatment were linked to internalized stigma amongst LAPs. Disfigurement produced feelings of shame when meeting HCWs or visiting the healthcare facility,[33, 52, 56] reinforced beliefs that leprosy cannot be healed,[9, 36] and caused feelings of being an 'eternal patient' after completing treatment.[54] Changed appearances also led to stigma in all other domains.

Stigma consequences at the health level

Stigma in the health domain produced health and psychological implications. Unpleasant actions from HCWs produced trauma,[56] stoked fears of universal hostility,[9] prompted further negative feelings towards HCWs,[28, 56] and reduced confidence in meeting HCWs, leading to worsening of disease.[56] Feelings of being an 'eternal patient' also left LAPs frustrated and pessimistic, causing them to avoid treatment[36] and abandon personal health.[54]

Economic: lack of motivation and opportunity

In the economic domain, enacted,[2, 9, 24, 26, 27, 38, 39, 45, 46, 50, 53, 54, 56] anticipated,[10] and internalized stigma[2, 9, 23, 27, 33, 37, 54] were linked to difficulty accessing jobs and to a reduced desire to be productive.

Stigma manifestation and its driving factors at the economic level

Enacted stigma in the economic domain manifested in a lack of job opportunities,[28, 31, 46, 47, 52] rejection when applying for jobs,[2] family members' suggestions to resign,[9] firings,[9, 23, 28] temporary suspensions until recovery,[39] exclusion and avoidance from co-workers,[34] awkwardness or avoidance from customers,[9, 38, 52] and lowered opportunities for financial support in business.[31] Physical appearance,[2, 52] concerns about contagiousness,[9] bank loan organizational policy,[31] and gender discrimination[46] were reasons for these experiences.

In the economic domain, only one anticipated stigma was found: fear of customers' avoidance.[10] Internalized stigma showed in declining drive towards productivity. LAPs sensed that they would be unable to do or find jobs,[27, 33] or would be unfit for jobs.[23, 37] They stopped working,[9] labelling themselves as fit only to beg[54] or become objects of charity.[2] Feelings of shame over working in daylight[27] also suggest internalized stigma. These internalizations were intensified by changes in appearance,[2, 27] disability,[23] decreased physical performance,[9, 33] and community stigma[23] including a local association between LAPs and chronic disease, wounds, begging, and poverty.[54]

Stigma consequences in the economic level

Stigma impacted the occupational, financial, social, mental, and health status of LAPs. They experienced loss of customers,[9, 52] bankruptcy,[34] mismatch between current and potential work,[23] and limitations on work (e.g. working only at night) in order to avoid being noticed.[27] Unemployment may cause poverty and criminal behaviour,[31] dependence on external aid,[23, 35] begging,[23, 46] deprioritizing treatment,[29] worsening health,[39] self-labelling as a permanent patient,[2] and decisions to reside in the place where LAP's were previously treated.[27] Psychologically, LAPs felt discouragement,[29, 34] self-doubt,[33, 37] lack of passion,[31] frustration,[35] and suicidal thoughts or attempts.[34, 47] Stigma in the economic domain also caused LAPs to conceal their disease from customers.[10]

Education: constraints in education access and development

Stigma manifestation and driving factors at the educational level

In the education domain, enacted stigma emerged in the form of insults,[9, 28, 29, 35] others' physical distance,[9, 34] gossip when a student's parents contracted leprosy,[47] and a school bus avoiding the area where many LAPs lived.[29] Some drivers mentioned were fear of contagion and presence of visible affected skin.[29]

LAPs felt anticipated stigma, such as fear of mockery or avoidance,[9] or fear of their child's being shunned by schoolmates.[29] Internalized stigma in the education domain manifested in feelings of shame when

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attending school.[29, 35] No studies clearly articulated the drivers of these behaviours. However, these felt stigma may be influenced by limited school options near many LAP's residential area.[29] This prevents an infected child from changing schools, so that they bear stigma throughout their education.[29] The importance of education in Indonesian society may also increase education-related stigma.[9]

Stigma consequences in the education level

Indonesian children who were infected with leprosy, or whose family members had leprosy, were reported to leave school temporarily or permanently.[9, 29–31, 34–36, 48, 54] Others continued their education but described feeling shame[23] or hiding their condition from school authorities.[29] Studies of stigma in the education domain focused entirely on school attendance and participation.

Public entitlements: 'distance' to market, transportation, and places of worship

Stigma manifestation and driving factors at the public entitlements level

Stigma towards LAPs decreases access to public facilities. LAPs experienced unfair treatment at small shop and markets, on public transportation, and in places of worship. At the small shop and market, some sellers refused to sell to the LAP[29, 30] and LAPs experienced staring.[29] LAPs were expelled from public transportation,[30, 48] and had difficulty accessing transportation from their residential area.[29] In places of worship, LAPs experienced exclusion and others' unwillingness to physically approach.[47, 48] Reasons cited for these forms of enacted stigma were physical appearance,[29] fear of transmission,[29] and local connotations attached to LAPs.[48]

Feelings of shame in public could reflect internalization of stigma. For example, studies mentioned embarrassment about appearing at places of worship.[29, 39] Reasons for this included physical changes such as darkened skin[39] and deformities.[29] Requirements that footwear be removed before praying also increased this hesitation, because of the possibility that a deformity in the foot might be seen.[29]

Stigma consequences at the public entitlements level

Suicidal thoughts was reported as a psychological impact of discrimination in public entitlements such as religious settings and public transportation.[47] Other consequences of stigma in this domain could not be found.

Coverage of stigma-affected domains

Supplement 6 (Coverage of domains impacted by stigma in included literature) shows coverage of the domains in the reviewed studies. This table reveals how extensive research has discussed the stigma affecting each domain LAPs lives in Indonesia.

Discussion

This review aimed to systematically retrieve qualitative evidence on leprosy-related stigma in Indonesia. Drawn from literature, the synthesis reveals seven domains of life penetrated by stigma in Indonesia: community, domestic, intimate relationship, health, economic, education, and public entitlements.

The 'community' domain was the most extensively studied of these domains in Indonesia. Stigma in the 'domestic', 'intimate relationship', 'health', and 'economic' domains had also received attention. Forms, drivers, and consequences of stigma were least-explored in the 'education' and 'public facilities' domains, indicating lack of concern, knowledge, and research. Internalized and perceived stigma were more difficult to identify in the literature relative to experienced stigma.

Stigma drivers were not always related to negative concepts associated with the disease concepts. Other factors, such as HCW visit and economic struggles were also found to induce stigma. These results imply that causes of stigma can be broad, including factors apparently unrelated to disease as well as seemingly positive ones. Thus, anti-stigma interventions for LAP must account for these broad influences. In terms of research, this would complement a previous global approach review.[3]

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This study recommends that efforts to reduce leprosy-related stigma in Indonesia must affect all seven domains. This is not only because the consequences of stigma in each domain were upsetting and at times violated human rights, but also because the above results show that events in one domain influenced others. For example, avoidance from HCWs (health domain), promoted feelings of social unworthiness in LAPs (community domain)[9] and avoiding health facilities increased risk factors for stigma in community interaction.[29]

The above may also be an addition to the recent Health and Stigma Discrimination Framework by Stangl *et al* (2019).[57] The stigmatization process may be fuelled not only by stigma manifestations and drivers, but also by stigma consequences and outcomes. Moreover, the findings of this study would mean that stigma across domains is interrelated, suggesting that anti-stigma programs should be performed simultaneously in all impacted domains.

These findings offer suggestions regarding the focus of anti-stigma programs in each domain. In community life, an open environment for LAPs to participate in meaningful practices, such as greetings, is needed. Cultural leaders may become involved in restructuring long-held local labels for LAPs. In the domestic domain, programs should focus on marital prospects, quality of marital life, family roles, and family cohesiveness. In the public domains, particularly healthcare services, programs should focus on preventing disability and improving HCW's manner. HCWs should be trained in sharing bad news and offering therapeutic communication to the LAP and families, especially during diagnosis. HCWs should also offer support to family members, caregivers, and key decision makers on how to care for and communicate with LAP during and after treatment. Finally, HCWs should ensure adequate rehabilitation after therapy is completed. In the economic domain, programs should focus on providing jobs and preventing job loss, including by building awareness among employers. Programs in the education domain should focus on creating friendly school environments for LAPs or family members of LAPs. Regarding public entitlements, access to the market, transit, and religious sites, as well as acceptance and equal treatment in these places, are required.

Qualitative studies in nine out of ten leprosy-endemic provinces (using 2017 data)[13] are scarce. This suggests the need for exploratory studies on the experiences of LAPs in at-risk populations. Such studies would increase contextual awareness for the larger public, practitioners, and researchers. Knowledge of stigma in the domains of education and public entitlements in Indonesia are needed. Views and experiences from children and key persons in the structural field, both private and government-owned, are also essential.

Limitations

There are several limitations on this study. Firstly, before it was conducted, systematic review was not registered on the relevant platform, though a search on registered reviews was carried out to prevent duplication. Secondly, in terms of study searching, only online databases and open-access literature were used. Next, some articles were obtained from local peer-reviewed journals that were not yet indexed in international databases. Grey literature, known to lack peer-reviewing, were also analysed. The authors did not exclude such literature in their search in order to obtain more knowledge and include more Indonesian language-written literature, which is difficult to find in international databases. To ensure all reviewed literature met quality requirements, authors performed quality checking with CASP. CASP is one of the most-used tools for quality appraisal[58] and is endorsed by Cochrane,[59] yet authors believed risk bias could still occur because the final essential criteria for deciding minimum required quality can be subjective.[60] Furthermore, CASP appeared less sensitive to validity than other tools such as ETQS and JBI. Finally, although risk of bias in the interpretation of meanings was minimized by the researchers, who were all Indonesian and used a consensus step to resolve discrepancies, it should be noted that text analysis of qualitative studies may still reflect differences in meaning relative to the original author of the article.

Conclusion

LAPs in Indonesia felt internalized, anticipated, and experienced stigma in both private and public domains. These domains included intimate relationship, domestic, neighbourhood, occupational, educational, healthcare, and other essential public settings. Stigma in the community domain, and experienced-type

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stigma, appeared most prominently in qualitative research conducted in Indonesia. Ten categories of stigma drivers and five areas of stigma consequences were found. Anti-stigma programs should address all domains, adapting to each one's unique challenges, including stigma generators specific to the local culture. Further research is needed to explore the stigma felt and experienced by LAPs, its determinants, and its consequences outside Java provinces, in education, and in public entitlements. Future studies may also find whether stigma experienced in one domain worsens feeling of stigmatization in others.

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Figure Legend:

Figure 1. PRISMA flow diagram depicting the study selection process

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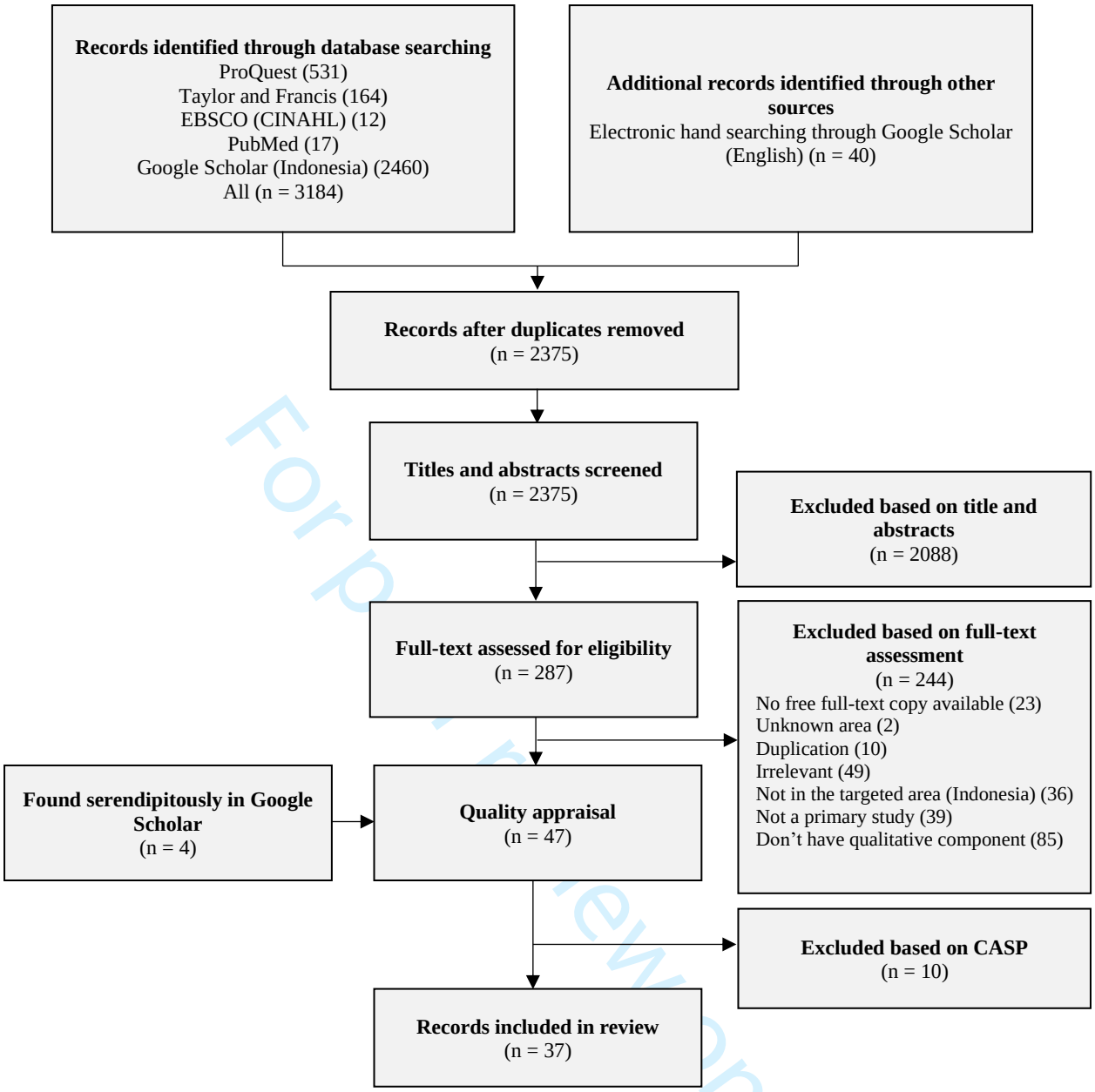


Figure 1. PRISMA flow diagram depicting the study selection process

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Supplement Material 1

Syntaxes for search

Databases	Syntaxes	Results
ProQuest Database	(("risk factor" OR "risk factors") OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR "quality of life" OR manifestation* OR outcome* OR satisfaction) AND (leprosy OR "Hansen's Disease" OR "Hansen Disease" OR leprae) AND (qualitative OR anthropolog* OR cultur* OR "focus group" OR "focus-group" OR ethnography OR "grounded theory" OR interview) AND Indonesia)	531 (Scholarly Journal Article) + 18 (Dissertation /Theses)
Taylor and Francis	(“risk factor*” OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR “quality of life” OR manifestation* OR outcome*) AND (leprosy OR “Hansen's Disease” OR “Hansen Disease” OR leprae) AND Indonesia AND (qualitative OR anthropolog* OR cultur* OR “focus group” OR “focus-group” OR ethnography OR “grounded theory” OR interview)	164 (Scholarly Journal Article)
EBSCO (CINAHL)	(“risk factor*” OR cause* OR determinant* OR stressor* OR belief* OR construct* OR practice* OR driv* OR facilitat* OR effect* OR influence* OR result* OR stigma OR “quality of life” OR manifestation* OR outcome*) AND (leprosy OR “Hansen's Disease” OR “Hansen Disease” OR leprae) AND Indonesia AND (qualitative OR anthropolog* OR cultur* OR “focus group” OR “focus-group” OR ethnography OR “grounded theory” OR interview)	12 (Scholarly Journal Article)
PubMed	#1“risk factor*”[tw] OR cause*[tw] OR determinant*[tw] OR stressor*[tw] OR belief*[tw] OR construct*[tw] OR practice*[tw] OR driv*[tw] OR facilitat*[tw] #2 impact*[tw] OR effect*[tw] OR influence*[tw] OR result*[tw] OR stigma[tw] OR “quality of life”[tw] OR manifestation*[tw] OR outcome*[tw] OR satisfaction[tw] #4 leprosy[tw] OR “Hansen's Disease”[tw] OR “Hansen Disease”[tw] OR leprae[tw] #5 Indonesia[tw] #6 qualitative[tw] OR anthropolog*[tw] OR cultur*[tw] OR “focus group”[tw] OR “focus-group”[tw] OR ethnography[tw] OR “grounded theory”[tw] OR interview[tw]	17 (Scholarly Journal Article)
Google Scholar (English)	leprosy AND qualitative AND (stigma OR "quality of life") AND ("east java" OR "west java" OR gorontalo OR sulawesi OR papua OR maluku)	40 (Electronic Hand Searching)
Google Scholar (Indonesia)	*kusta AND kualitatif ("faktor risiko" OR penyebab OR determinan OR "sumber daya" OR dampak OR pengaruh OR stigma OR "kualitas hidup" OR sulawesi OR gorontalo OR maluku OR papua OR jawa OR indonesia)*	2460 (All type of publication)

Supplement Material 2

Critical Appraisal Skills Programme (CASP) score for the reviewed studies

No	Study (first author, year)	CASP										
		Was there a clear statement of the aims of the research?	Is qualitative methodology appropriate?	Was the research design appropriate to address the aims of the research?	Was the recruitment strategy appropriate to the aims of the research?	Was the data collected in a way that addressed the research issue?	Has the relationship between researcher and participants been adequately considered?	Have ethical issues been taken into consideration?	Was the data analysis sufficiently rigorous?	Is there a clear statement of findings?	How valuable is the research?	CASP Score
1	Al Bana, 2014[30]	1	1	1	1	1	0	1	1	1	1	9
2	Arisal, 2020[46]	1	1	1	1	Can't tell	1	0	1	1	1	8
3	Armaijn, 2019[47]	1	1	1	1	Can't tell	0	0	Can't tell	1	1	6
4	Aulia, 2019[56]	1	1	1	1	Can't tell	1	0	0	1	1	7
5	Dadun, 2016[31]	1	1	1	1	1	0	1	1	1	1	9
6	Dary, 2017[25]	1	1	1	1	Can't tell	0	1	1	1	1	8
7	Fajar, 2002[48]	1	1	Can't tell	1	1	0	0	Can't tell	1	1	6
8	Hidayat, 2020[50]	1	1	1	1	1	0	1	1	1	Can't tell	8
9	Huda, 2011[49]	1	1	1	1	1	0	0	1	1	1	8
10	Indow, 2019[53]	1	1	1	1	1	0	1	1	1	1	9
11	Jatimi, 2020[32]	1	1	1	1	1	0	1	1	Can't tell	1	8
12	Jufriyanto, 2020[26]	1	1	1	1	1	0	1	1	1	1	9
13	Kusharnanto, 2013[33]	1	1	1	1	1	Can't tell	1	1	1	1	9
14	Lusli, 2015[2]	1	1	1	1	1	Can't tell	1	1	1	1	9
15	Lusli, 2016[35]	1	1	1	1	1	1	1	1	1	1	10
16	Lusli, 2017[34]	1	1	1	1	1	1	Can't tell	1	1	1	9
17	Marbaits, 2012[36]	1	1	1	1	1	0	0	1	1	1	8
18	Nainggolan, 2017[37]	1	1	1	1	1	0	0	Can't tell	1	1	7
19	Najmuddin, 2013[54]	1	1	1	1	1	0	1	1	1	1	9
20	Peters, 2013[9]	1	1	1	1	1	Can't tell	1	1	1	1	9
21	Peters, 2014[10]	1	1	1	1	1	1	1	1	1	1	10
22	Peters, 2016[38]	1	1	1	0	1	0	0	Can't tell	1	1	6
23	Pribadi, 2018[39]	1	1	1	1	Can't tell	0	Can't tell	1	1	Can't tell	6
24	Rai , 2020[51]	1	1	1	1	1	1	1	1	1	1	10
25	Rai , 2020[40]	1	1	1	1	1	1	1	1	1	1	10
26	Romadhon, 2020[41]	1	1	1	1	1	1	1	1	1	1	10
27	Saing, 2016[42]	1	1	1	Can't tell	Can't tell	0	0	1	1	1	6
28	Sandi, 2018[28]	1	1	1	1	1	Can't tell	1	1	1	1	9
29	Schuller, 2010[43]	1	1	1	1	1	Can't tell	Can't tell	1	1	1	8
30	Siregar, 2018[24]	1	1	1	1	Can't tell	0	1	1	1	1	8
31	Sodik, 2016[44]	1	1	1	1	0	0	0	0	1	1	6
32	Susanto, 2017[55]	1	1	1	1	1	0	1	1	1	1	9
33	Tarigan, 2019[52]	1	1	1	1	1	0	Cant tell	Cant tell	1	1	7
34	Verayanti, 2016[23]	1	1	1	1	1	0	0	1	1	1	8
35	Wahyuni, 2017[29]	1	1	1	Can't tell	1	0	0	1	1	1	7
36	Widodo, 2018[45]	1	1	1	1	1	0	0	0	1	1	7
37	Yudanagara, 2020[27]	1	1	1	1	1	0	Can't tell	1	1	1	8

Supplement Material 3

Description of stigma manifestations towards LAPs in Indonesia

Enacted
Avoidance[2,8, 22–27,30,31,33,39–42,49,52,53]
Rejection in greetings, handshake or physical touch[9, 27–29, 31, 40, 44, 53]
Reluctance to talk[24, 45, 50]
Absences of invitations to gatherings[29, 43–45]
Restriction to participate[2]
Negative comment, gossip, or mock[23, 24, 56, 29, 31, 33, 35, 47, 51, 52, 55]
Spitting on property and nose was covered[27]
Covering nose[27]
Starring[24]
Rejection to live in the same neighbourhood[41]
Expulsion[48, 50]
Unfair treatment due to social background[46, 51]
Recitation of a specific prayer to reject transmission[49]
Rejection to marry or to build romantic partner[28–30, 37, 43, 47, 49]
Acceptance and giving suggestion to divorce[35, 43]
Prohibition of touching family member[34]
Concealment of diagnosis by family members[40, 48]
The view as a family disgrace[23]
Family rejection even though has been cured[37]
Command not to take a bath[28]
Utensils, clothes, bed separation[48]
Suggestion to leave the house[25]
Shortage of job[31, 45, 47, 52]
Rejection from job[2]
Alienation at work[24, 35]
Suspension to work[39]
Dismissal from work[9, 23]
Refusal to be in the same body of water (e.g. river, water at rice field) [9, 29]
Rejection or hesitation to buy items sold by LAP[9, 29, 34, 38, 52]
Hesitation, rejection to help, or delayed treatment from healthcare provider[28, 31, 56]
Hurtful comments at school[9, 33, 34, 52]
Physical distance[35, 40, 47, 48, 50]
Absence of transportation at the residential area where LAP lived[29]
Rejection at the market[30]
Rejection from public transportation driver and passengers[27, 47, 48]
Anticipated
Fear of avoidance[9, 10, 24, 35, 36, 38, 47, 51, 52]
Fear of insults, humiliation, or negative comments[2, 9, 10, 24, 36, 43]
Fear people would reject to meet[10, 29, 35]
Fear of infecting others[25, 31]
Worry people will notice or suspect something wrong[10, 35, 47, 48]
Fear if interactions will make people will feel offended[27]
Worry the deceased body of LAP would not be buried at the public cemetery[30]
Fear family members will afraid of the disease[53]
Fear to be excluded by the family and neighbours[40, 47, 53]
Worry of being rejected to take care of family[10, 35]
Fear romantic partner will leave[31]
Fear difficulty to have a romantic partner[31]
Internalized
Feeling ashamed of self[9, 23, 56, 24, 28, 29, 32, 33, 35, 40, 55]
Seeing self as an abnormal or different[2, 29, 32, 43]
Seeing self as dirty, ugly, disgusting, unclean[27, 29–31, 35]
Seeing self as an eternal-patient, will not be healed or normal[2, 9, 27, 30, 34, 36, 51, 54]
Seeing self as a danger for other[35]
Feeling no one want to marry[29]
Seeing self a poor person and fit to be a beggar[54]
Seeing self as a family troublemaker or burden[32, 34, 43]
Seeing self as an object of charity[2, 29, 43]
Seeing self as having a dangerous disease[54]
Seeing a stamp of ‘leprosy’ at self[27]
Feeling inferior[9, 46]
Feeling people will not be friendly anymore[9]
Feeling as a cursed person even after cured[37]
Feeling unworthy[37]
Feeling will not be able to get a job or work again[27, 33]
Feeling people are afraid of self[27]

Supplement Material 4

Description of drivers of stigma towards LAPs in Indonesia

Drivers	Description	
Physical factors[2, 9, 33, 35, 37–40, 42, 43, 47, 50, 23, 51–54, 56, 25–30, 32]	Physical appearance due to deformity or a darkening skin, unpleasant physical odour, declining physical function, decreasing performance due to physical weakening, continuous pain after cured.	Related with the disease progress
Fear of contagion[2,7,8,22–24,26–28,31, 34,35,39,43,45,47,52,53,55]	Fear of fast transmission from breathing the same air, immersing in the same body of water, shaking hands, eating food prepared by the person affected, carrying the deceased body of a person affected, and using the same personal object such as glasses, towels, clothes with the person affected.	
Hereditary factor[2, 36, 43, 49, 50, 54]	Believe that leprosy can be inherited in the family line.	
Local connotation for LAP[23, 43, 48, 54]	Local connotation directed to the leprosy affected persons (e.g *'Nibamparekangji'* in South Sulawesi that has a meaning 'to ignore or treat them like they do not exist or *'Judam'* in East Java that means 'ugly and hated by people').	
Religious and/or mystical reason[22,24,31,32,34,45–49,50,53,55]	Believe that leprosy is caused by sin, curse, punishment on bad deeds, evil spirit influence, or spell.	
Inability to perform social role[32, 34, 35, 43, 46, 52, 54]	Lack or no opportunity to perform previous routine or responsibility in and for the family, continuous failure in relationship, job, school.	
Factors related to social and health assistance[9, 35–37, 48, 56]	Home visits from health care worker *[considered positive approach, yet could still drive stigma]*, exhaustion in caring for patients, care provider communication style, and an inadequate social rehabilitation.	
Beliefs around prognosis[34, 36, 42, 45, 49, 54]	Believe leprosy as a long-term disease, a high fatality disease, and an incurable disease.	
Residential area[29, 41]	Believe that the residential area where persons-affected live is a dangerous zone or by living there people will also get stigmatized.	
Social background[34, 46, 51]	Ethnic, religion, gender, or economic background.	Not related with the disease progress

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Supplement Material 5

Description of stigma consequences on LAPs in Indonesia

Social behaviour
Self-isolation[7,8,22,26–28,30,31,45,47,48–51]
Avoiding new people, home visits, and social activities[2, 9, 27, 28, 35, 44, 47]
Apathy to others[36]
Hiding activities from family members[38]
Difficulty to marry, build romantic relationship or friendship[2, 27, 30, 37, 43]
Estranged or damaged relationship with significant others[9, 31, 34, 37]
Being passive at home[2, 28, 35, 36]
Changing place to live[23, 25, 27, 33, 37, 46, 52]
Hiding face or body parts when going outside[29, 33, 35, 47]
Loss of social image[40]
Limited inter-island movement[41]
Being used for political and economic purpose[41, 46]
Psychological
Sadness[8,25–28,36,38,45,49,24,51,52]
Frustration[9, 27, 28, 33, 35–37, 52, 53]
Despair[9, 28, 35]
Guilt[32, 43]
Felling lonely[2, 25, 34, 35]
Feeling nsafe[47]
Anger or easy to be irritated[37, 39]
Inability to be angry[29]
Pessimism[9, 23, 35, 36, 51]
Loss of self-confidence[28, 35, 44]
Feeling unfulfilled[37]
Feeling useless[29]
Being ashamed of self (for a long time period)[9, 27, 32, 38, 45, 46, 52]
Self-disappointment[29, 35, 52]
Spiritual distress (e.g. questioning God or stopping routine religious activity)[28, 37]
Loss of passion to live[31, 36–38]
Suicidal thinking[9, 10, 25, 27, 29, 30, 35, 36, 47]
Suicidal attempt[36]
Occupational/financial
Resigning from work[9, 24, 35]
Difficulty to find job[2, 31, 33, 45]
Losing customer[9, 38, 52]
Preference to work at night[27]
Feeling shy to find a proper job[23, 33]
Weak financial condition[31, 34, 46]
Dependence on aid to sustain life (e.g. becoming a beggar or leaning on government's aid)[23, 31, 46, 54]
Health
Concealing, lying or denying the diagnoses[2,7,8,26,30,33,36,37,48,49,52]
Avoiding healthcare facility or provider[9, 29, 36, 40, 41, 47, 48, 54, 56]
Abandoning the sign of leprosy[36, 48]
Seeking treatment secretly[48]
Seeking traditional healer[40, 48]
Medication noncompliance[40, 48]
Lack of family support in treatment[48]
Absence in health education[54]
Unhygienic practice of self-care[28]
Worsening disease[39, 40, 56]
Weakened physical health[36, 39]
Excessive use of drugs[52]
Neglecting general health[54]
Self-mutilation[10]
Education
Stop attending school[9, 29, 31, 33–35, 52]
Difficulty to access school bus[29]
Having teacher who refuses to live on the same island with the students[41]

Supplement Material 6

Coverage of domain impacted by stigma in included literature

| Themes | Literature | | |

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2 & attached document "Abstract Checklist"
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3 (paragraph 1-3)
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 3 (paragraph 3)
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 3
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 3 (search strategy)
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 4 (search strategy)
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4 (study selection)
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 4 (data extraction and analysis)
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 4 (data extraction and analysis)
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 4 (data extraction and analysis)
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4 (study selection), Supplement material 2 (appraisal)
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	N/A
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 4 (study selection)
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 5 (synthesis)
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 5 (synthesis)
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 5 (data extraction and analysis)
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 4 (search results)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Page 4,5,and table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplement material 2 (appraisal)
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	N/A
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Supplement material 2 (appraisal)
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 12-13
	23b	Discuss any limitations of the evidence included in the review.	Page 13 (limitation)
	23c	Discuss any limitations of the review processes used.	Page 13 (limitation)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	23d	Discuss implications of the results for practice, policy, and future research.	Page 13 (paragraph 6)
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 13 (limitation)
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 14 (funding and acknowledgement)
Competing interests	26	Declare any competing interests of review authors.	Page 14
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 14 (data sharing statement)

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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